

FACULTY
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Charles University

MASTER THESIS

Bc. Eduard Ješko

Vibrational optical activity of biomolecules

Institute of Physics of Charles University

Supervisor of the master thesis:	prof. RNDr. Vladimír Baumruk, DrSc.
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Author: Bc. Eduard Ješko

Department: Institute of Physics of Charles University

Supervisor: prof. RNDr. Vladimír Baumruk, DrSc., Institute of Physics of Charles University

Abstract: The thesis aims at the study of conformation of a dimethyl tartrate molecule using the methods of vibrational optical activity (VOA), namely vibrational circular dichroism (VCD) and Raman optical activity (ROA). Based on the theoretical background of both VOA methods and current state of research of the studied molecule there was a sample of dimethyl tartrate dissolved in different solvents and its properties were measured using VCD and ROA spectrometers. In addition to the experiment, *ab initio* calculations were carried out in order to compare calculated and experimental spectra. Based on the comparison, the possible conformations present in water solution of the studied molecule are described in detail.

Keywords: vibrational circular dichroism, Raman optical activity, dimethyl tartrate, *ab initio* spectral simulations

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Preface

The field of vibrational optical activity (VOA), consisting mainly of vibrational circular dichroism (VCD) and Raman optical activity (ROA) has undergone a vast transformation during the turn of the millennia. Originally a specific area of research which was practiced by a few scientists, it evolved into a new field of spectroscopy and is now practiced by a large number of scientists around the globe. The main factors that enabled this transition were the rapid development of commercially available instrumentation and software for routine measurements and quantum calculations of VOA. The reason for this development was the need of chemists to control and characterize chirality of the molecular solutions they synthetized. Moreover, chirality became critical especially in the pharmaceutical industry, the most successful drugs being single enantiomers. In the United States, regulations requiring proof of the absolute configuration of every new drug molecule were established. Up to these days, scientists are still discovering the possibilities of VOA to give high quality information on chirality in molecular science.

On the other hand, the application of VOA is not only in molecular chirality. For biomolecules, the role of chirality is not centered on the specification of absolute conformation, but more on the solution-state conformation in their native solutions. Extreme sensitivity of VOA was discovered for many types of biomolecules, i.e. peptides, amino-acids, sugars, etc., but also viruses and bacteria. Moreover, attention has moved to the analysis of the human genome: to the understanding of the structure and roles of the molecules therein.

The key feature that enables VOA to probe conformation in new ways is the combination of the three dimensional stereo-sensitivity of optical activity and the precise structural sensitivity of vibrational spectroscopy. It is in fact typically a traditional spectrum, like vibrational absorption or Raman scattering together with the additional VOA spectrum that corroborate the novel spectral information.

One of the main goals of this thesis was to probe the conformation of a molecule called dimethyl tartrate which is a derivative of the well known tartaric acid molecule. This molecule has been proven to be conformationally very flexible, thus it represents a difficult yet scientifically interesting topic to study. The thesis is organized into six chapters. The first chapter contains a brief overview of VCD, ROA and of the studied molecules. The second chapter presents the theory behind both VOA methods. In the third chapter the basic computation procedures of VCD and ROA are introduced. The fourth chapter contains information about the current state of research of the studied molecule as well as the objectives of the presented work. In the fifth chapter the experimental methods used for measuring the sample are described. The sixth chapter contains results and discussion of measurements and theoretical calculations. Moreover it contains a comparison of measured and experimental spectra in order to describe the possible conformations present in water solution of the studied molecule which is one of the main goals of this work. A list of conclusions is presented in the last chapter of this work.

1. Introduction

1.1 Fundamentals of vibrational optical activity

Vibrational optical activity (VOA) is a form of natural optical activity which dates back to the nineteenth century. It is known that the first observations of optical activity such as the rotation of the plane of linearly polarized light, called optical rotation or the differential absorption of left and right circularly polarized light, called electronic circular dichroism are caused by electronic transitions in molecules. VOA is a form of natural optical activity which is based on vibrational transitions in molecules, rather than electronic transitions.

VOA can be defined as the difference of interaction of left and right circularly polarized light with a molecule which is caused by a vibrational transition in the molecule. Such a broad definition allows for large variety of spectroscopic techniques. The two most important and most frequently used are vibrational circular dichroism (VCD) and Raman optical activity (ROA). VCD is an absorption technique while ROA is a scattering technique. VCD is typically measured for infrared radiation with wavenumbers between 800 cm^{-1} and 4000 cm^{-1}. On the other hand, ROA can be measured from 50 cm^{-1} up to 2500 cm^{-1}, in certain cases may be extended up 4000 cm^{-1} [1].

1.1.1 Vibrational circular dichroism

VCD is defined by the following relation [2]:

$$\Delta A(\nu) = A_L(\nu) - A_R(\nu) \tag{1.1}$$

where A_L and A_R is the absorbance for left and right circularly polarized light. The way VCD is defined here (i.e. "left" minus "right") is consistent with the definition used for electronic circular dichroism (ECD), which is based on electronic transitions in the molecule.

For an achiral molecule with random orientation of the molecules, the difference ΔA is zero. The definition of VCD is illustrated in Figure 1.1 for a molecule which is undergoing a transition from the zeroth to the first vibrational level of the ground electronic state.

In order for VCD to be non-zero, either the molecule has to be chiral or it has to be in a chiral molecular environment. For chiral molecules, the two enantiomers yield ΔA which for each frequency has the same magnitude, but opposite sign. Thus VCD spectra of the two enantiomers are mirror images with respect to the frequency (or wavenumber) axis.

When Beer's law is obeyed [3], the VCD intensities are defined by relationship:

$$\Delta\varepsilon(\nu) = \varepsilon_L(\nu) - \varepsilon_R(\nu) = \Delta A(\nu)/(cl) \tag{1.2}$$

where $\Delta\varepsilon$ is called differential molar extinction coefficient, c is the concentration and l is the pathlength. This definition of VCD does not depend on the choice of the sampling variables such as pathlength and concentration, thus is useful when comparing experimental and calculated VCD spectra. For experimental measurements it is useful to define a quantity called anisotropy ratio:

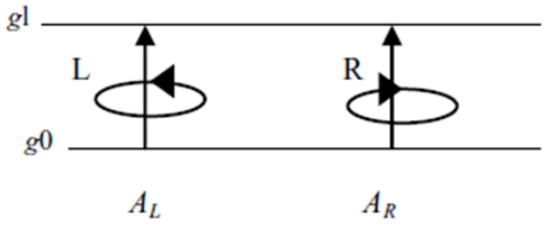


Fig. 1.1: Diagram illustrating energy levels and VCD on a molecule in the ground electronic state which is undergoing a vibrational transition from the zeroth to the first vibrational level (adapted from [2]).

$$g = \frac{\varepsilon_L - \varepsilon_R}{\varepsilon_L + \varepsilon_R}, \tag{1.3}$$

which reflects the intensity ratio between vibrational absorption (VA) and VCD signals. VCD anisotropy ratios are typically $<10^{-4}$ [4].

1.1.2 Raman optical activity

ROA is defined as the difference in chiral molecular systems for right and left circularly polarized incident and/or scattered light [2]:

$$\Delta I(\nu) = I^R(\nu) - I^L(\nu). \tag{1.4}$$

ROA is a two-photon process in comparison with the complementary one photon absorption based VCD. Nonetheless both ROA and VCD deal with identical vibrational transitions. Circular polarization ROA has four forms, as opposed to VCD, which has only one. The respective energy level diagrams are shown in Figure 1.2.

The original and the oldest form of ROA is today called incident circular polarization (ICP) ROA. In this case, the incident laser is modulated between right and left circular polarization states, while the Raman scattered light intensity is detected at a fixed linear or an unpolarized radiation state.

The second form of ROA is termed scattered circular polarization (SCP) ROA. Here, fixed linear or unpolarized incident laser radiation is used and the difference in intensity of right and left circularly polarized Raman scattered light is measured.

The third form of ROA is in-phase dual circular polarization (DCP_I) ROA. In this case, the polarization states of both the incident and scattered radiation are switched synchronously between RCP and LCP states.

Finally, the fourth form of ROA is termed out-of-phase dual circular polarization (DCP_{II}) ROA. Here, the polarization states of both the incident and scattered radiation are switched oppositely between LCP and RCP states. The definitions of these forms of ROA for any vibrational transition involving a normal mode between the states ev and ev' are given by the following expressions for each form of ROA [2]:

$$\text{ICP} - \text{ROA} \quad \Delta I_\alpha = \Delta I_\alpha^R - \Delta I_\alpha^L \tag{1.5}$$

$$\text{SCP} - \text{ROA} \quad \Delta I^\alpha = \Delta I_R^\alpha - \Delta I_L^\alpha \tag{1.6}$$

$$\text{DCP}_\text{I} - \text{ROA} \quad \Delta I_I = \Delta I_R^R - \Delta I_L^L \tag{1.7}$$

$$\text{DCP}_\text{II} - \text{ROA} \quad \Delta I_{II} = \Delta I_L^R - \Delta I_R^L. \tag{1.8}$$

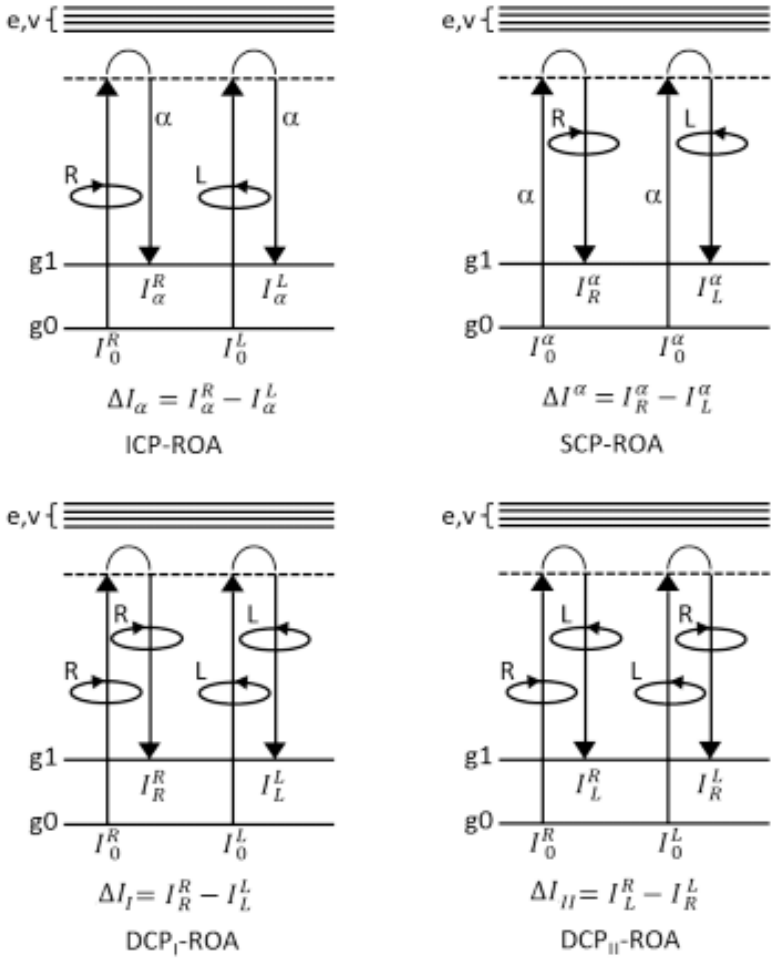


Fig. 1.2: A diagram illustrating energy levels and the definition of ROA on a molecule in the ground electronic state which is undergoing a vibrational transition from the zeroth to the first vibrational level for different modulation schemes (image adapted from [2]).

.

Moreover, ROA can be defined using different scattering geometries, such as forward (0°), right-angle (90°) or backward (180°) setup and distinct frequencies of incident radiation [2].

For experimental measurements it is useful to define a dimensionless quantity called circular intensity difference (CID)

$$\Delta = \frac{I^R - I^L}{I^R + I^L}, \tag{1.9}$$

which reflects the intensity ratio between ROA and Raman signals. This quantity differs for different ROA scattering geometries [2].

1.2 Chirality of biological molecules

Chirality is a property of asymmetry important in different branches of science.

The word 'chiral' is derived from the Greek word 'kheir' meaning hand. Chiral objects are characterized by the absence of symmetry plane, center and rotation-reflection axis. Perhaps the easiest way to illustrate this property is on the example of human hands. The left hand is the mirror image of the right hand, however, they are not identical, see Figure 1.3. More precisely, the object is called chiral if it cannot be superimposed on its mirror image. Most of the biomolecules can be characterized by this property and typically only one of the possible forms (enantiomeric or diastereoisomeric) is biologically active.

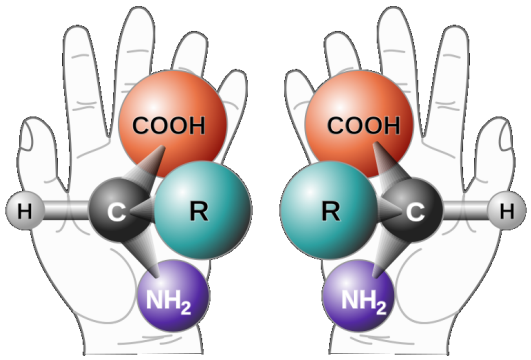


Fig. 1.3: Two enantiomers of a generic amino acid that are chiral. Image adapted from [5].

Important topological concept associated with chirality are chiral stereocenters (i.e. an asymmetric carbon atoms, which have bonds with four different substituents). For example, the proteolytic enzyme chymotrypsin has 251 stereocenters with 2^{251} possible stereoisomers. Nevertheless, only one stereoisomer is utilized in living organisms [6]. Small biomolecules which are the main field of focus in this work are often described by the general R, S system [7] which represents the information about the chirality of the molecule. In larger molecules such as peptides/proteins or nucleic acids the repetition of subuntis (for example L-amino acid residues in case of peptides/proteins or D-sugars in case of nucleic acids) are associated with chiral segments of secondary structures (sheets, helices) and are responsible for chiroptical spectra. Biologically important processes of biomolecules is nowadays dynamically researched topic which is one of the main fields of application of chiroptical spectroscopy.

1.3 Tartaric acid and its derivatives

The main subject of study of this work is a small biomolecule, which is a derivative of tartaric acid called dimethyl tartrate (both enantiomers). More precisely, it is an ester of tartaric acid. In the following sections a brief

introduction about tartaric acid which is at the moment a well documented molecule is presented. After that an introduction to the less documented dimethyl tartrate (more precisely to both its enantiomers) will be given.

1.3.1 History

Tartaric acid is a white crystalline organic acid that occurs naturally in many plants, most notably in grapes. It has played an important role in the discovery of chirality. This property was first observed in 1832 by Jean Baptiste Biot, who observed its ability to rotate polarized light. Louis Pasteur further continued Biots research by investigating chiral ammonium sodium tartrate crystals. He was the first to produce a pure sample of levotartaric acid by sorting the differently shaped crystals under a magnifying glass.

1.3.2 Stereochemistry and chemical properties

The naturally occurring form of the acid is (2R, 3R)-(+)-tartaric acid or simply called levotartaric acid. The mirror image of this substance can be produced artificially and it is called (2S, 3S)-(–)-tartaric acid or dextrotartaric acid. Both of these molecules are optically active. A diastereomer of the both acids called it (2R, 3S) mesotartaric acid can be also produced artificially and is optically inactive. The projections of these substances are shown in Figure 1.4. Table 1.1 shows the basic chemical properties of these substances.

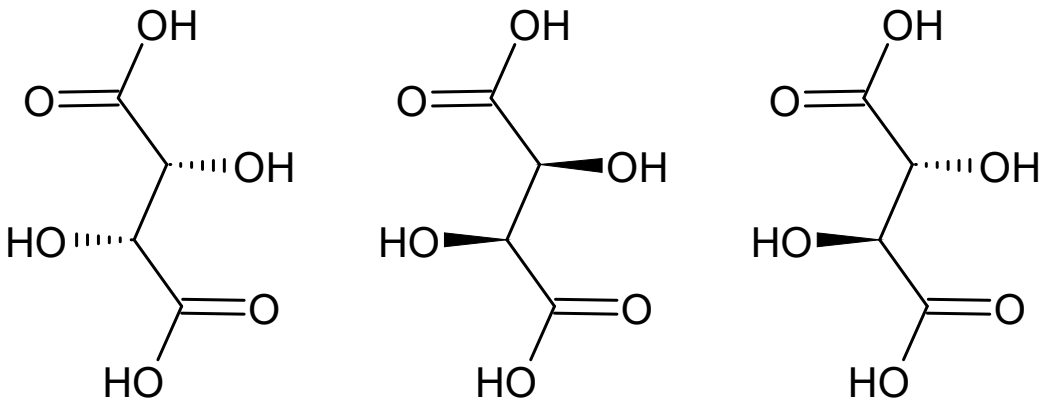


Fig. 1.4: Image of levotartaric acid (left), dextrotartaric acid (middle) and mesotartaric acid (right).

1.3.3 Occurrence and application

As stated earlier, tartaric acid occurs in plants, mainly grapes. That is the reason why it is recognizable to wine drinkers as the source of "wine diamonds" (see Figure 1.5). These so called diamonds may occur on the cork or at the bottom of the bottle. From a medical point of view they are harmless although nowadays their creation is being prevented by a process called cold stabilization. In the

Tab. 1.1: Basic chemical properties of tartaric acid isomeres [8].

Acid type	Levotartaric	Dextrotartaric	Mesotartaric
Molar mass (g/mol)	150.09	150.09	150.09
Density (g/ml)	1.79	1.79	1.79
Melting point (°C)	171 to 174	171 to 174	146 to 148
Solubility in water (kg/l)	1.33	1.33	1.25
Acidity (pK_{a_1}) at 25 °C	2.89	2.89	3.22
Acidity (pK_{a_2}) at 25 °C	4.4	4.4	4.85

past these tartrates from the inside of wine barrels were collected for industrial usage.

Tartaric acid and its derivatives are widely used mainly in the food and pharmaceutical industry [9]. During fermentation they play an important role chemically, lowering the pH in order to prevent the creation of unwanted bacteria and acting as a preservative after fermentation. As a food additive, they can be used as an antioxidant or as an emulsifier.



Fig. 1.5: "Wine diamonds"

Usage of this substance in the pharmaceutical industry is common, for example, in combination with citric acid, it is used in the production of effervescent salts to improve the taste of oral medications. One of the derivatives of tartaric acid is used in cough syrup as an expectorant.

This acid is also widely used in the industry as a chelating agent. It is able to chelate metal ions such as calcium and magnesium. That is the reason it is used in the farming and metal industries as a chelating agent for complexing micronutrients in soil fertilizer and for cleaning metal surfaces consisting of aluminium, copper, iron, and alloys of these metals, respectively [8].

1.3.4 Dimethyl tartrate

Tartaric acid and its derivatives play an important role in organic and pharmaceutical chemistry. Dialkyl tartrates are a category of tartaric acid derivatives which have wide applications in both academic laboratories and chemical industries. They can be used as good catalysts for asymmetric synthesis, and they are also chiral intermediates for several pharmaceuticals and agrochemicals.

The main field of focus of this work were two molecules, (2R, 3R)-(+)-dimethyl tartrate. For this molecule throughout of this work the abbreviation L-DMT will be used. Secondly, its enantiomer (2S, 3S)-(−)-dimethyl tartrate was investigated. For this molecule abbreviation D-DMT will be used. Both of these molecule have

two chiral centers and are optically active. When referring to both molecules, we will use abbreviation DMT (which stands for dimethyl tartrate). The molecules consist of 22 atoms and their molar mass is 178.14 g/mol. The projections of these molecules are shown in Figure 1.6. The molecules are soluble in water as well as in less polar solvents. The reasons and objectives of study of these two enantiomers will be presented in Section 4.



Fig. 1.6: Image of the (2R,3R)-(+)-dimethyl tartrate (L-DMT) (left) and of (2S,3S)-(–)- dimethyl tartrate (D-DMT) (right).

2. Theory of vibrational optical activity

2.1 Born-Oppenheimer approximation

In order to predict the vibrational spectra of a molecule, we need to know the wavefunctions and energies of the states of the molecule, predicted by the Schrödinger equation

$$\hat{H}\Psi_i(\mathbf{r},\mathbf{R}) = E_i\Psi_i(\mathbf{r},\mathbf{R}) \tag{2.1}$$

where $\Psi_i(\mathbf{r},\mathbf{R})$ and E_i are the wavefunction and the energy of the ith state which depend on the sets of Cartesian coordinates of the electrons r and nuclei R, $\hat{H}$ is the molecular Hamiltonian

$$\hat{H} = \hat{T}_e + \hat{T}_n + V_{ee} + V_{en} + V_{nn} \tag{2.2}$$

where $\hat{T}_e$ is the electronic kinetic energy, $\hat{T}_n$ is the nuclear kinetic energy, V_{ee} is the potential energy of electron-electron repulsion, V_{en} is the potential energy of electron-nucleus attraction and V_{nn} is the potential energy of nucleus-nucleus repulsion. While the Schrödinger equation can be solved analytically for the atom of hydrogen, the analytical solution of the Schrödinger equation for a molecule with multiple electrons and nuclei is mathematically impossible. Consequently, the wavefunctions and energies of a molecule are obtained using the Born-Oppenheimer (BO) approximation, as follows.

Since the nuclei are much heavier than electrons (approximately 1000 times) we assume that the electrons are moving in the field of fixed nuclei and that the fast moving electrons create an effective field for the nuclei. Due to this fact the Schrödinger equation is solved for the electrons with the fixed nuclei

$$\left(\hat{H} - \hat{T}_n\right)\Psi_i(\mathbf{r},\mathbf{R}) = W_i\Psi_i(\mathbf{r},\mathbf{R}) \tag{2.3}$$

where $\Psi_i(\mathbf{r},\mathbf{R})$ defines the electron distribution of the ith electronic state at nuclear geometry $\mathbf{R}$ and $W_i(\mathbf{R})$ is the potential energy surface (PES). Next, the nuclear Schrödinger equation is solved

$$\left(W_i(\mathbf{R}) + \hat{T}_n\right)\chi_{ij}(\mathbf{R}) = E_{ij}\chi_{ij}(\mathbf{R}) \tag{2.4}$$

where χ_{ij} is the nuclear wavefunction of the jth state. Finally, the BO wavefunction is obtained by combining the electronic and nuclear wavefunction

$$\Psi_{ij}(\mathbf{r},\mathbf{R}) = \Psi_i(r,R)\chi_{ij}(\mathbf{r},\mathbf{R}). \tag{2.5}$$

The energy of this wavefunction is E_{ij}. The BO approximation is considered one of the main methods used in quantum chemistry in order to solve the Schrödinger equation for a molecule.

2.2 Harmonic approximation

Now we consider a molecule in lowest lying energy state, named ground state, G. For a molecule with N nuclei there are 3N nuclear coordinates. Let us assume that for one nuclear geometry the potential energy W_G is a minimum on the potential energy surface (PES), and that nuclear vibration is limited to geometries close to this minimum. This geometry, $\mathbf{R^0}$, is termed the equilibrium molecular geometry. Now by expanding the PES around the equilibrium in a Taylor series we get the following expression:

$$W_G(\mathbf{R}) = W_G(\mathbf{R^0}) + \sum_{\lambda\alpha} \left(\frac{\partial W_G(\mathbf{R})}{\partial X_{\lambda\alpha}} \right)_{R=R^0} X_{\lambda\alpha} + \\ + \frac{1}{2} \sum_{\substack{\lambda,\alpha \\ \lambda',\alpha'}} \left(\frac{\partial^2 W_G(\mathbf{R})}{\partial X_{\lambda\alpha} \partial X_{\lambda'\alpha'}} \right)_{R=R^0} X_{\lambda\alpha} X_{\lambda'\alpha'} + \dots \tag{2.6}$$

where $X_{\lambda\alpha} = (\mathbf{R}_{\lambda\alpha} - \mathbf{R}^0_{\lambda\alpha})$ are the Cartesian displacement coordinates, λ defines the nucleus and α defines the Cartesian axes, X, Y, and Z. We can set term $W_G(\mathbf{R^0})$ equal zero. Since $\mathbf{R^0}$ is the minimum of the PES, the second term of Equation 2.6 is zero. By neglecting the higher terms of the Taylor series we get the harmonic approximation as:

$$W_G(R) = \frac{1}{2} \sum_{\substack{\lambda,\alpha \\ \lambda',\alpha'}} \left(\frac{\partial^2 W_G(\mathbf{R})}{\partial X_{\lambda\alpha} \partial X_{\lambda'\alpha'}} \right)_{R=R^0} X_{\lambda} X_{\alpha} \tag{2.7}$$

where $(\partial^2 W_G(\mathbf{R})/(\partial X_{\lambda\alpha} \partial X_{\lambda'\alpha'}))_0$ is the Hessian matrix of the PES. In order to solve the nuclear Schrödinger equation a transformation of the coordinates to 3N new nuclear coordinated Q_a is needed. These new coordinates are called normal coordinates and they are linear combinations of the Cartesian displacement coordinates.

$$Q_a = \sum_{\lambda,\alpha} S_{\lambda,a} X_{\lambda} \tag{2.8}$$

Now with the help of Equation 2.8 we transform Equation 2.7 into

$$W_G(\mathbf{R}) = \frac{1}{2} \sum_{a=1}^{3N} k_a Q_a^2 \tag{2.9}$$

where $k_a = \partial^2 W_G(\mathbf{R})/\partial Q_a^2$. Also the nuclear kinetic energy T_n is transformed by Equation 2.8 into

$$T_n = \frac{1}{2} \sum_{a} \dot{Q_a}^2 \tag{2.10}$$

Finally we obtain the nuclear Schrödinger equation in terms of the normal coordinates

$$\left(\frac{1}{2} \sum_{a=1}^{3N} k_a Q_a^2 + \frac{1}{2} \sum_{a=1}^{3N} \dot{Q_a}^2 \right) \chi_{Gj}(Q_a) = E_{Gj} \chi_{Gj}(Q_a) \tag{2.11}$$

Since the Hamiltonian in Equation 2.11 is the sum of the Hamiltonians of 3N one dimensional harmonic oscillators, the wavefunction χ_{Gj} and energy E_{Gj} are given by

$$\chi_{Gj}(Q_a) = \prod_{a=1}^{3N} \chi_{v_a}(Q_a) \tag{2.12}$$

$$E_{Gj} = \sum_{a=1}^{3N} \left(v_a + \frac{1}{2} \right) h\nu_a \tag{2.13}$$

where $\chi_{va}(Q_a)$ is the wavefunction of a one dimensional harmonic oscillator of force constant k_i and quantum number v_a and ν_a is the frequency of this harmonic oscillator, where $\nu_a = (1/2\pi)\sqrt{k_a}$. Out of 3N normal coordinates Q_a, 3N-6 describe nuclear motions in which the relative positions of atoms changes; these coordinates are the vibrational normal modes. The rest can be associated with translational and rotational motion of the molecule. Since the energy of the molecule does not change when the molecule translates or rotates, the force constants and frequencies of the six translational and rotational normal modes are all zero. Only the vibrational normal modes have nonzero force constants and frequencies [4].

2.3 Theory of VCD

VCD studies the changes of vibrational states of a molecule by its interaction with infrared light. Since the sizes of molecules studied in this work are much smaller than the wavelength of infrared light we assume uniform electric and magnetic field over the molecule. The electromagnetic perturbation of a molecule exposed to uniform electric and magnetic fields $\mathbf{E}$ and $\mathbf{H}$ is

$$H'(\mathbf{E}, \mathbf{H}) = -\boldsymbol{\mu} \cdot \mathbf{E} - \boldsymbol{m} \cdot \mathbf{H} \tag{2.14}$$

where $\boldsymbol{\mu}$ and $\boldsymbol{m}$ are the electric and magnetic dipole moments of the molecule.

In order to predict VA and VCD spectra, time dependent perturbation theory (TDPT)[10] is used. To determine vibrational absorption spectra only the electric dipole contribution to the perturbation of the molecule has to be included (called the electric dipole approximation). Using this approximation, the molar extinction coefficient spectrum can be written as a sum over all normal mode transitions a of a sample [2]

$$\varepsilon(\nu) = \frac{8\pi^3 N\nu}{(2.303)3000hc} \sum_a D_a(i \to f) f'_a(\nu) \tag{2.15}$$

where i and f are the initial and final states in the exctitation $i \to f$, $f'_a(\nu)$ is the shape of the absorption band which is most frequently Lorentzian (applies also to VCD) and $D_a(i \to f)$ is the dipole strength of transition $i \to$ f of ath normal mode:

$$D_a(i \to f) = |\langle i \,|\boldsymbol{\mu}|\, f \rangle_a|^2 \tag{2.16}$$

where $|\langle i \,|\boldsymbol{\mu}|\, f \rangle_a|$ is the electric dipole transition moment of the transition $i \to f$ of ath normal mode.

In addition, to predict VCD spectra both electric and magnetic dipole contribution to H' have to be included. Then nonzero VCD is predicted [2]

$$\Delta\varepsilon(\nu) = \frac{32\pi^3 N\nu}{(2.303)3000hc} \sum_a R_a(i \to f) f'_a(\nu) \tag{2.17}$$

$R(i \to f)_a$ is the rotational strength of excitation $i \to f$ of ath normal mode which has the following form

$$R_a(i \to f) = \mathrm{Im}\left[\langle i \left|\boldsymbol{\mu}\right| f\rangle_a \cdot \langle i \left|\boldsymbol{m}\right| f\rangle_a\right] \tag{2.18}$$

where $\langle i \left|\boldsymbol{\mu}\right| f\rangle_a$ is the electric dipole transition moment (real quantity) and $\langle i \left|\boldsymbol{m}\right| f\rangle_a$ (imaginary quantity) is the magnetic dipole transition moment of ath normal mode. The electric dipole transition moment is calculated within the BO approximation and harmonic approximation.

However, within the BO approximation for molecules with non-degenerate ground electronic states (which can by applied to all neutral organic molecules), the magnetic dipole transition moment of fundamental transition is zero, because the magnetic dipole moment of a nondegenerate electronic state is zero at all nuclear geometries. That is the reason why a more accurate vibrational wavefunctions has to be constructed than the BO wavefunctions. By mixing BO wavefunctions of the ground electronic state G with those of the excited states E and by utilizing perturbation theory, nonzero magnetic dipole transition moment can be obtained. This method was developed by Stephens and more detail can be found in [11].

In general, the anisotropy ratio (introduced in Section 1.1.1) for VCD in terms of dipole and rotational strength as

$$g = \frac{4R(\nu)}{D(\nu)}. \tag{2.19}$$

When calculating VCD spectra, molecular properties such as atomic axial tensor and atomic polar tensors need to be determined in order to calculate the rotational strength in Equation 2.18. More details may be found [4]. This method of calculation has been implemented into Gaussian software already in 1998 [12] and is currently considered as a standard method for VCD calculations [13].

2.4 Theory of ROA

ROA calculations are much more complex in comparison to VCD calculations. ROA studies the vibrational states of a molecule by its interaction with visible light. If the frequency of the incident light is smaller than the frequency of the electronic absorption of the molecule, we may use the far-from-resonance Raman scattering approximation. However, in order to compute ROA spectra, it is necessary to move beyond the dipole approximation and include not only the electric dipole contribution, but also the magnetic dipole and electric quadrupole contributions. Tensors, which are responsible for the generation of the ROA signal are the electric dipole-electric dipole polarizability tensor

$$\alpha_{\alpha\beta} = \frac{2}{\hbar} \sum_{i \neq f} \left(\frac{\nu_{if}}{\nu_{if}^2 - \nu^2} \right) \mathrm{Re} \left(\langle f \left| \hat{\mu}_\alpha \right| i \rangle \langle i \left| \hat{\mu}_\beta \right| f \rangle \right) \tag{2.20}$$

the magnetic dipole–electric dipole ROA tensor

$$G'_{\alpha\beta} = -\frac{2}{\hbar} \sum_{i \neq f} \left(\frac{\nu}{\nu_{if}^2 - \nu^2} \right) \mathrm{Im} \left(\langle f \left| \hat{\mu}_\alpha \right| i \rangle \langle i \left| \hat{m}_\beta \right| f \rangle \right) \tag{2.21}$$

and electric quadrupole–electric dipole ROA tensor

$$A_{\alpha\beta\gamma} = \frac{2}{\hbar} \sum_{i \neq f} \left(\frac{\nu_{if}}{\nu_{if}^2 - \nu^2} \right) \mathrm{Re} (\langle f | \hat{\mu}_\alpha | i \rangle \langle i | \hat{\theta}_{\beta\gamma} | f \rangle) \tag{2.22}$$

where $\nu_{if} = \nu_i - \nu_f$ is the angular frequency difference between the initial state i and final state f, ν is the frequency of the incident light, and $\hat{\theta}_\alpha$ is the electric quadrupole term.

ROA intensities from samples of randomly oriented molecules can be expressed in terms five tensor invariants, which have the following form:

$$\beta(\alpha)^2 = \frac{1}{2} \left(3\alpha_{\alpha\beta}\alpha_{\alpha\beta} - \alpha_{\alpha\alpha}\alpha_{\beta\beta} \right) \tag{2.23}$$

$$\beta(G')^2 = \frac{1}{2} \left(3\alpha_{\alpha\beta}G'_{\alpha\beta} - \alpha_{\alpha\alpha}G'_{\beta\beta} \right) \tag{2.24}$$

$$\beta(A)^2 = \frac{1}{2} \nu \alpha_{\alpha\beta} \varepsilon_{\alpha\gamma\delta} A_{\gamma\delta\beta} \tag{2.25}$$

$$\alpha = \frac{1}{3} \alpha_{\alpha\alpha} \quad \text{and} \quad G' = \frac{1}{3} G'_{\alpha\alpha} \tag{2.26}$$

With the help of the above derived invariants the intensity of the Raman scattering ($I^R + I^L$) and ROA ($I^R - I^L$) can be expressed [14]

$$I^R + I^L = 4K \left(D_1 \alpha^2 + D_2 \beta(\alpha)^2 \right) \tag{2.27}$$

$$I^R - I^L = \frac{8K}{c} \left(D_3 \alpha G' + D_4 \beta(G')^2 + D_5 \beta(A)^2 \right) \tag{2.28}$$

where c is the speed of light and K is a constant. In general, from Equations 2.27 and 2.28 the dimensionless CID can be expressed which was introduced in Section 1.1.2 as

$$\Delta = \frac{\frac{8K}{c} \left(D_3 \alpha G' + D_4 \beta(G')^2 + D_5 \beta(A)^2 \right)}{4K \left(D_1 \alpha^2 + D_2 \beta(\alpha)^2 \right)} \tag{2.29}$$

Coefficients $D_1 - D_5$ are dependent on the configuration of the ROA experiment and are shown in Table 2.1.

In order to obtain simulated spectra, we have to sum over all normal modes a and the intensity of Raman/ROA spectra at each point is given by

$$S(\nu) = \sum_a I_a^{RAM/ROA} f'_a(\nu) \tag{2.30}$$

where f'_a is Lorentzian shaped function.

Tab. 2.1: Coefficients $D_1 - D_5$ for different modulation schemes and scattering geometries which in Equations 2.27, 2.28 and 2.29. The subscripts in the modulation column mean: u-unpolarized, p-parallel polarization, d-perpendicular polarization, *-magic angle polarization. Table adapted from [15].

scattering geometry	modulation schema	D_1	D_2	D_3	D_4	D_5
0°	ICP_u	45	7	90	2	-2
	SCP_u	45	7	90	2	-2
	DCP_I	45	1	90	2	-2
	DCP_{II}	0	6	0	0	0
90°	ICP_p	45/2	7/2	45/2	7/2	1/2
	ICP_d	0	3	0	3	-1
	ICP^*	45/3	10/3	45/3	10/3	0
	SCP_p	45/2	7/2	45/2	7/2	1/2
	SCP_d	0	3	0	3	-1
	SCP^*	45/3	10/3	45/3	10/3	0
	DCP_I	45/4	13/4	45/2	13/2	-1/2
	DCP_{II}	45/4	13/4	0	0	0
180°	ICP_u	45	7	0	12	4
	SCP_u	45	7	0	12	4
	DCP_I	0	6	0	12	4
	DCP_{II}	45	1	0	0	0

3. *Ab initio* methods and computational procedures

Calculation of VOA spectra consists of several steps. At this level, VCD and ROA spectra can be treated on an equivalent basis.

3.1 Basic principles of *ab initio* calculations

For multi-electron systems it is not possible to solve the Schrödinger Equation 2.3 analytically. Therefore *Ab initio* methods have been developed which solve this equation numerically. Hatree-Fock (HF) theory is the most known *ab initio* method [16] . Although the computational expense of this method is the least expensive of all the *ab initio* methods, the accuracy of this method is not sufficient due to its incomplete description of the correlation between motions of electrons. Therefore more sophisticated methods, which include electron correlation, such as perturbation theory or configuration interaction have been developed [16]. The disadvantages of these methods are expensive computational resources and that they are not practical for large systems.

The most used technique nowadays, which is used exclusively also in this work, is the density functional theory (DFT) [17]. The basis of this method is the Hohenberg-Kohn theorem [18], which demonstrates the existence of a unique functional that determines the ground state energy and density, although the theorem does not provide the form of this functional. Nevertheless, significant improvements have been made in order to develop approximate functionals. One of the most popular functionals is the B3LYP functional [19], which is used in this work. It has been shown, that DFT can give results of similar quality using similar basis sets comparable to more sophisticated methods at a computational cost of the HF method. Generally, DFT calculations improve systematically on increasing the size and flexibility of the basis functions used to construct the functionals. For example the 6-311++G** basis set contains polarization and diffuse functions and gives sufficient results compared to computational time. Overview of DFT functionals and basis sets is available in [4].

3.2 Computational procedures

3.2.1 Geometry optimization

The first step is to determine the geometry of the molecule. Next, for further calculation in case of DFT, a functional and basis set is chosen. Usually the most reliable functionals are hybrid functionals such as B3LYP or B3PW91, although due to the inherent nature of DFT, no one functional is consistently the most reliable. More functionals should be tried out in order to obtain the best agreement with experimental results. The 6-31G* basis set is the smallest basis set that provides reasonable results. For smaller molecules and powerful computers larger basis sets such as TZ2P or cc-pVTZ improve the accuracy of the computed spectra, although at much larger computational costs.

Next, geometry optimization is performed. Geometry optimization attempts to locate minimum on the PES. At minimum, the first derivatives of the energy with respect to the Cartesian displacement coordinates are zero. The gradient indicates the direction along the PES in which the energy decreases most rapidly from the current point as well as the steepness of the slope. Next the value of the second derivative of the energy with respect to Cartesian displacement coordinates, updating the matrix of force constants (Hessian matrix), which specifies the curvature of the surface is computed. These calculation are performed within the harmonic approximation. More details about geometry optimization may be found in [20]. For conformational flexible molecules more than one of the minimum energy geometries on the PES are populated. Then different energy minima which correspond to different conformations of the molecule are included in the final VA/VCD resp. Raman/ROA spectra with respect to their Boltzmann distribution [21].

3.2.2 Solvent models

In some cases, consideration of the influence of the solvent on the molecule is important and needs to be taken into account. More precisely, polar solvents (such as water) have a substantial effect on the calculated spectra. There are multiple ways how the solvent interaction may be taken into account.

Molecules of the solvent may be taken into account explicitly, in other words the molecules of water are directly placed around the studied molecule. The disadvantages of this model is, that the studied molecule is influenced in a unique way and different solvent conformations should be calculated. For larger systems and due to substantial computational time of quantum chemical calculations, this method is not ideal.

Another easier method how to incorporate the solvent into calculations is the implicit solvent method. One of the simplest levels is the polarizable continuum model (D-PCM), which surrounds the solute molecule with a conforming shell with the dielectric constant of the solvent. More frequently used model is the conductor like PCM (C-PCM) in which the continuum is conductor-like similar to COSMO solvation model [22]. This mimics the effect of the solvent on the dissolved molecules and can alter the vibrational frequencies of some modes and possibly change the energy ordering of the conformers. This C-PCM solvent models was used in this work. More details about implicint solvent models can be found in [22].

3.2.3 Force fields, vibrational frequencies, vibrational intensities and bandshapes

After performing conformational analysis and geometry optimization, the vibrational frequency calculation for each of the minimum energy geometries obtained from the conformational analysis is calculated. Next, Hessian matrix (Equation 2.7), atomic polar and axial tensors which are necessary to calculate vibrational frequencies, dipole (Equation 2.16) and rotational strengths (Equation 2.18) which yield VA (Equation 2.15) and VCD (Equation 2.17) intensities.

In case of Raman scattering and ROA intensities, Raman and ROA tensors and the corresponding invariants are calculated which were presented in Section 2.4. Incident laser radiation frequency has to be chosen before calculation. Depending on the experimental setup the scattering geometry and the modulation schema has to be selected. More details regarding intensity calculation may be found in [2].

The experimental bandshapes of VA/VCD resp. Raman/ROA spectra are most frequently Lorentzian, calculated spectra are plotted using Lorentzian lineshape function

$$f'_a(\nu) = \frac{1}{\pi\gamma_a} \frac{\gamma_a^2}{(\nu - \nu_a)^2 + \gamma_a^2} \tag{3.1}$$

where γ_a is the half width at its half height, which can be selected. This expression of Lorentzian bandshape may be appointed into Equations 2.15 resp. 2.17 for VA resp. VCD intensities and into Equation 2.30 Raman/ROA intensities.

4. Objectives

The main reason of measurements and theoretical calculation is to study the conformation of DMT.

To date, multiple studies of this molecule were performed from both computational and experimental side of view. However, the studies from the computational side of view considered DMT in a solvent of low polarity. The reasons for this is that the solvent effects on the computed spectra are much smaller in comparison with polar solvents such as water.

In case of ROA, to our knowledge, only one attempt was performed to measure DMT in H_2O [26]. In Figure 4.1 Raman and ROA spectra of L-DMT in H_2O is shown. It can be seen that since the measurement was performed in 1970s, low signal/noise ratio is present in the ROA spectrum. Back then the instrumentation of ROA was not as advance as it is nowadays. Today ROA spectrometers allow measurements in broader spectral region and substantial improvement of signal/noise is achieved. To our knowledge, no measurements were made in different solvents expect for H_2O. One of the the main goals of this work was to improve measured ROA spectrum in H_2O and to investigate solvent effects also in less polar solvents.

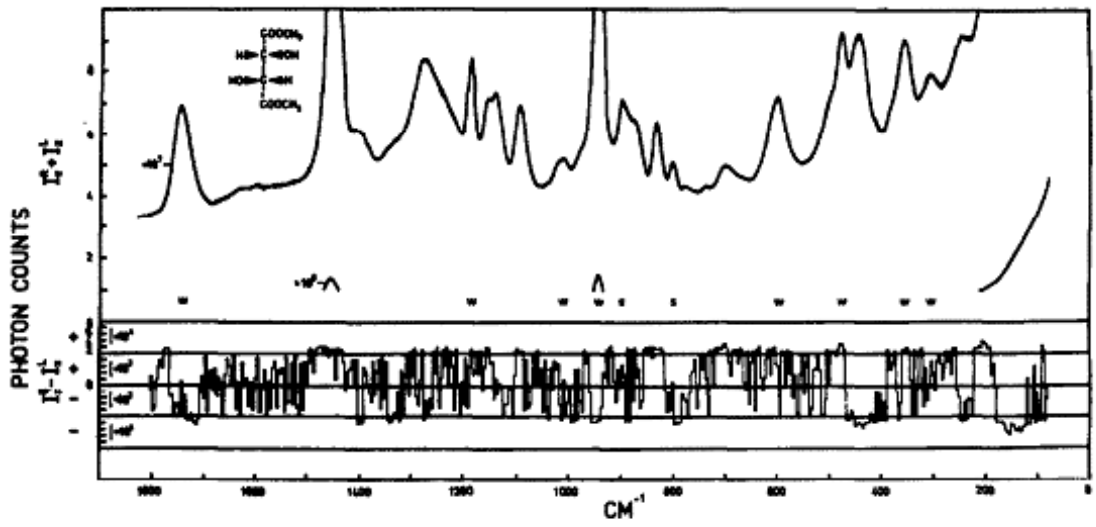


Fig. 4.1: Raman (bottom) and ROA (top) spectra of L-DMT in H_2O (spectrum taken from [26]).

In case of VCD, almost every measurement was carried out in solvents of low polarity, mostly CCl_4 [9, 27, 28, 29, 30]. In case of water environment, to our knowledge, the most recent VCD spectrum of D-DMT in D_2O is shown in Figure 4.2. Almost zero VCD signal is observed in this figure. According to article [9] also VCD spectrum in H_2O was measured, however with very little success. That is the reason why the main focus of this work is the water environment.

Moreover, we also performed measurements in less polar environments, such as $CHCl_3$ or DMSO mainly for two reasons: firstly, in order to reproduce the results presented in different papers and to test the newly acquired VCD instrument at

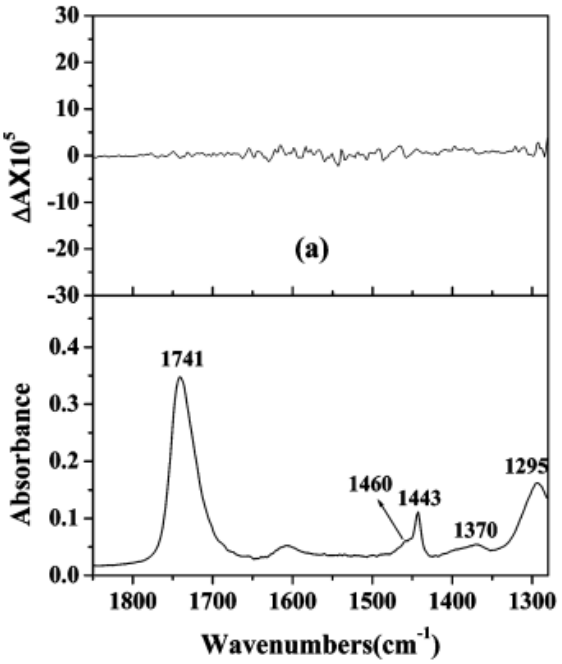


Fig. 4.2: VA (bottom) and VCD (top) of D-DMT in D_2O (spectrum taken from [9]).

the Institute of Physics and secondly, to observe changes in spectral bands which could help us to determine the conformation of the molecule.

The main goal of this work and also of VOA in general is to determine the conformation of the molecule with the help of theoretical simulations. This is achieved by constructing a theoretical spectrum that compares favorably with experimental data. By *ab Initio* calculations the conformations of DMT were studied in multiple works [9, 27], although, there were only solvents of low polarity considered. *Ab initio* calculations of the crystal structure of DMT were also carried out, more details can be found in [23]. To our knowledge, no calculations were carried out considering water, which is the most natural environment for biomolecules. That is the reason, why we carried out calculations by considering water. Results followed by discussion from measurements and calculations are presented in Chapter 6.

5. Experimental methods

5.1 Samples

Two samples were measured in this work, D-DMT(242942–25G) and L-DMT(163457–25G) both from Sigma-Aldrich chemical company.

5.2 ROA experiments

Raman and ROA spectra were acquired on the commercial spectrometer Chiral $RAMAN$-$2X^{TM}$ from Biotools (SCP-Biotools) at the Institute of Organic Chemistry and Biochemistry and on a home-built ROA spectrometer at the Institute of Physics at Charles University (ICP-CU). The experimental setup of these spectrometers is slightly different. The main difference is the usage of the SCP and ICP modulation schemes (see Figure 1.2) and Section 1.1.2.

5.2.1 ROA spectrometer at the Institute of Physics

This spectrometer uses the ICP schema where the modulation of polarization of the incident laser beam is used. The ROA spectrum is measured as the difference in Raman scattering intensity for right circularly polarized (RCP) minus left circularly polarized incident radiation. For changes of polarization between RCP and LCP the instrument is equipped by an electro-optic modulator (EOM). In the case of this spectrometer the EOM is a longitudinal Pockels cell based on a potassium dideuteriophosphate crystal. A 514.5 nm argon laser (Coherent, Innova) is the source of laser radiation. A holographic SuperNotch-Plus filter (Kaiser Optical Systems) is used to suppress Rayleigh scattering. The instrument is equipped by a fast stigmatic spectrograph HoloSpec HS-f/1.4 (Kaiser Optical Systems) which allows us to measure with the spectral resolution of 6.5 cm^{-1}. Different gratings are availabe for the argon laser in order to measure in different spectral regions: HSG-514.5-LF which allows us to measure in the ∼100–2370 cm^{-1} spectral region and a HSG-514.5-HF which allows us to measure in the 2270-4510 cm^{-1} spectral region. A CCD detector (1340x100 pixels, Roper Scientific) ensures light acquisition. Measured spectra were calibrated by a neon lamp. This instrument works in backscattering geometry which is currently considered the best experimental strategy [31]. Detailed description of this instrument is available in [15].

The main disadvantage of SCP experimental setup is the presence of the so called "flicker noise" [32] which might be caused for example by temporal instabilities of the laser beam, density fluctuations in the sample or by the presence of moving dust particles through the laser beam. This flicker noise may be suppressed by using the SCP modulation schema.

5.2.2 Chiral$RAMAN$-$2X^{TM}$

This instrument is based on the concept of W.Hug [33] and uses the SCP configuration schema. In comparison with the above mentioned instrument this

device measures the RCP and LCP scattered components simultaneously, i.e. ROA spectrum is measured as the difference in Raman scattering intensity for right circularly polarized minus left circularly polarized scattered radiation. This configuration allows us to get rid of the flicker noise. Substantial increase of measurement speed is also achieved. The instrument is equipped by a 2W green diode laser (OPUS 532) with an excitation wavelength of 532 nm and enables measurements in the ∼10–2430 cm^{-1} spectral region. The edge filter (532-nm RazorEdge, LP03-532RE-25, Semrock, Inc.; SCP-P B) is used to suppress Rayleigh scattering. Toluene spectra are used for calibration. This spectrometer works in the backscattering geometry similarly as the above mentioned instrument. More details are available in [34].

5.2.3 Experiment preparation and post processing

All samples were prepared at 1.6 M concentration. Next, they were filtered through 0.20 μm filter (Millipore Ultrafree-MC) which allows us to get rid of unwanted particles (dust for example).

For measurements on the ICP-CU spectrometer it was important to set the power of the laser. It was adjusted to the intensity of the Raman signal (750 mW in case of H_2O solutions, 500 mW in case of DMSO and $CHCl_3$ solutions.) The exposition time was set to 2 s. The total accumulation time for one sample was $\sim$ 22 hours. For measurements in the spectrometer a quartz 5x5 mm seized cuvette (Hellma) was used which was filled by 100 μl of sample. All samples were measured at 25 °C.

In case of SCP-Biotools the power of the laser was also adjusted to the intensity of Raman signal (1 W in case of H_2O solutions, 200 mW in case of DMSO solutions and 250 mW for $CHCl_3$ solutions). The exposition time was set to 1.03 s. Acquisition time for each sample was $\sim$ 24h. For measurements a 3x4 mm seized microcuvette (Starna Scientific Ltd.) was used which was filled by 80 μl of sample. All samples were measured at 25°C.

The obtained spectra from ICP-CU spectrometer went through post processing procedures such as clean field or Fourier transform in order to get rid of artifacts and reduce the noise in the spectra [15]. These procedures were not necessary to perform on the obtained spectra from the SCP-Biotools. Next, the GRAMS/AI software was used to treat spectra. First, the solvent spectrum was subtracted from the measured Raman spectrum and linear baseline correction was done for Raman as well as for ROA spectra. In case when samples exhibited fluorescence background, baseline correction was performed using higher order polynomials. Since both enantiomers of DMT were available, the presented ROA spectra of D-enantiomers were obtained by subtracting the ROA spectra of L-enantiomers from their corresponding ROA spectrum of the D-enantiomers and then dividing by 2 in order to improve the quality of the spectra. The presented level of artifact spectra (LoA) were obtained by summing the spectra of the D and L enantiomer and dividing the summed spectrum by 2.

Exact wavenumbers to the bands of the Raman spectrum were assigned by using the second derivative of the spectra. Bands of the ROA spectra were labeled by finding the maxima/minima of each band since due to high noise the use of second derivative method was not possible.

The presented Raman spectra were multiplied by a normalization factor, so that the band of highest intensity at ∼946 cm^{-1} had the $I_R + I_L$ factor equal to one. The corresponding ROA spectra were multiplied by the same factor.

5.3 VCD experiment

5.3.1 Chiral*IR*-2X^{TM}

VCD spectra were measured on a commercial spectrometer Chiral*IR*-2X^{TM} from Biotools at the Institute of Physics at Charles University. The core of the instrument is a Fourier transform dual source SiC [35] infrared spectrometer which precedes the polarization-modulation stage of the instrument. The creation or selective detection of left (LCP) and right circularly polarized (RCP) radiation is carried out with the combination of a polarizer and dual photo elastic modulator (Hinds instrument), one before and one after the sample [36]. Usage of two photo elastic modulators leads to substantial suppression of absorption artifact. After the infrared radiation passes through the sample, it is focused on a liquid nitrogen-cooled MCT (HgCdTe) detector. More details of the instrument is available at [34].

5.3.2 Experiment preparation and post processing

All samples were prepared at 1.6 M concentration. This concentration was proven to be ideal since it is the highest possible concentration of DMT where the absorbance of the sample is below one.

All of the measurments in different solvents were carried out in the 8 μm CaF_2 Biotools cuvette. The absorbtion of the cuvette allowed us to measure VCD spectra from 1070 cm^{-1} to higher wavenumbers. The VCD spectra were recored at a 4 cm^{-1} resolution. The cuvette was filled by 8 μl of sample. 18 blocks with 4000 scans per block were measured for each sample and the Blackman-Harris apodization function was used. This meant a collection time around ∼22 hours for each sample. All the measurements were carried out at 20 °C . However the actual temperature inside of the spectrometer was around 26 °C.

After the measurement the obtained VCD spectra were processed by the GRAMS/AI software. First, the solvent and water vapor spectrum was subtracted from the measured VA spectrum. The VCD spectrum of the solvent was also subtracted from the corresponding solution spectra in order to suppress possible experimental artifacts. Linear baseline correction was done for VA as well as for VCD spectra.

Since both enantiomers of DMT were available, labeled VCD spectra of D-enantiomers were obtained by subtracting the VCD spectra of L-enantiomers from their corresponding VCD of the D-enantiomers and then dividing by 2 in order to improve the quality of the spectra. The presented level of artifact (LoA) spectra were obtained by summing the spectra of the D and L enantiomer and dividing the summed spectrum by 2.

Exact wavenumbers to the bands of the VA spectrum were assigned by using the second derivative of the spectra. Bands of the VCD spectra were labeled by finding the maxima/minima of each band.

The presented VA spectra were multiplied by a normalization factor, so that the most intense band of the C=O vibration had the absorbance equal to one. Next, the corresponding VCD bands were multiplied by the same factor.

6. Results and discussion

6.1 Determination of the ideal assembling of the CaF_2 cuvette used on the VCD spectrometer

In order to perform measurements on the newly acquired VCD spectrometer, it was essential to adjust the gear used on this spectrometer.

Our task was to determine the ideal assembling of the VCD cuvette produced by the company *Biotools*. As the cuvette consists of two parts (shown in Figure 6.1) there are more possible ways how to assemble it. Our goal was to assemble the cuvette so that we obtain VCD spectrum with a baseline as straight as possible.

Measurement

According to the company, it was a 8 μm thick cuvette with a diameter d=25.6 mm made from CaF_2. The cuvette was filled by 8 μl of distilled water.

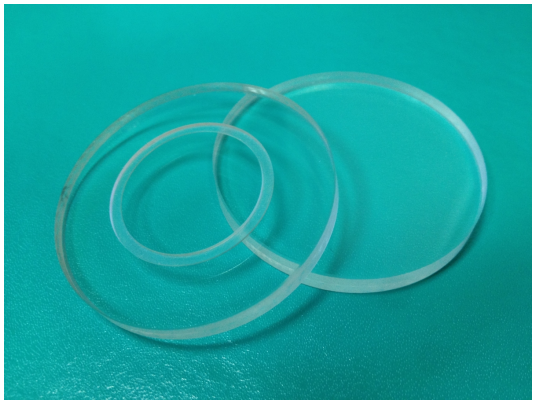


Fig. 6.1: CaF_2 cuvette

The measurement and spectral analysis was carried out in a spectral range from 1200-2000 cm^{-1} with 2000 scans were measured for each experiment.

Next the cuvette was assembled in different ways. A fixed position of the bottom glass was held and the top glass of the cuvette was rotated in the range from 0-180° with a step of 20°. A mark was made on the bottom glass of the cuvette in order to put the whole cuvette in the cuvette holder always at the same orientation. An assumption was made, that in the interval 180-360° the cuvette would behave in a symmetrical fashion but in order to prove this assumption measurements in orientations 220° and 320° were carried out. In Figure 6.2 VCD spectra of different positions of the top glass can be observed. Since lots of noise is present in the VCD spectra, fifth order polynomial fit was done in order to observe better the spectral profile.

It can be seen that the straightest VCD spectrum is around 20° and 100°. By further examination we conclude, that by rotating the cuvette by approx. 90° same spectrum can be obtained.

Orientation at 20° was selected for further examination. A more accurate orientation was searched in order to obtain an even more straight VCD spectrum. Besides position 20°, positions 10°, 15° and 30° were measured. In Figure 6.3 the raw and fitted VCD spectra in these orientations are shown.

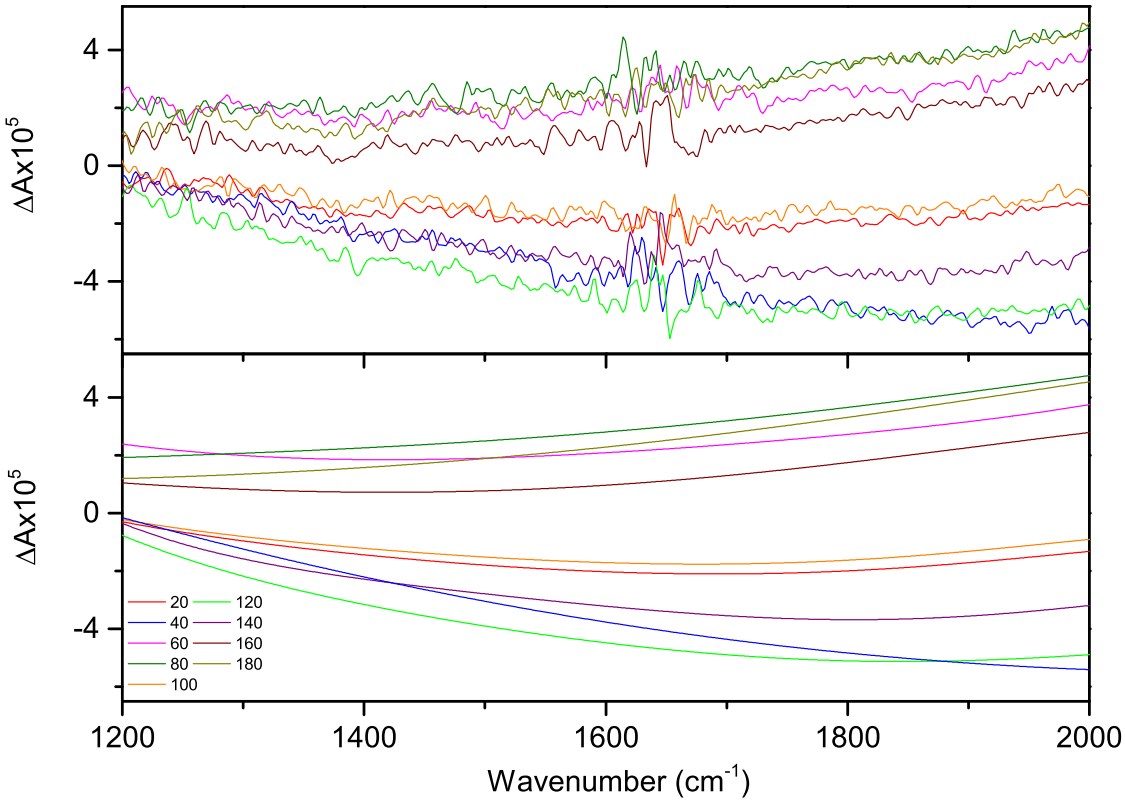


Fig. 6.2: Measured and fitted VCD spectra for different orientation of the top glass of the cuvette (the numbers stand for rotation of the top glass of the cuvette in degrees, for better clarity of the graph orientations 220° and 320° were not included).

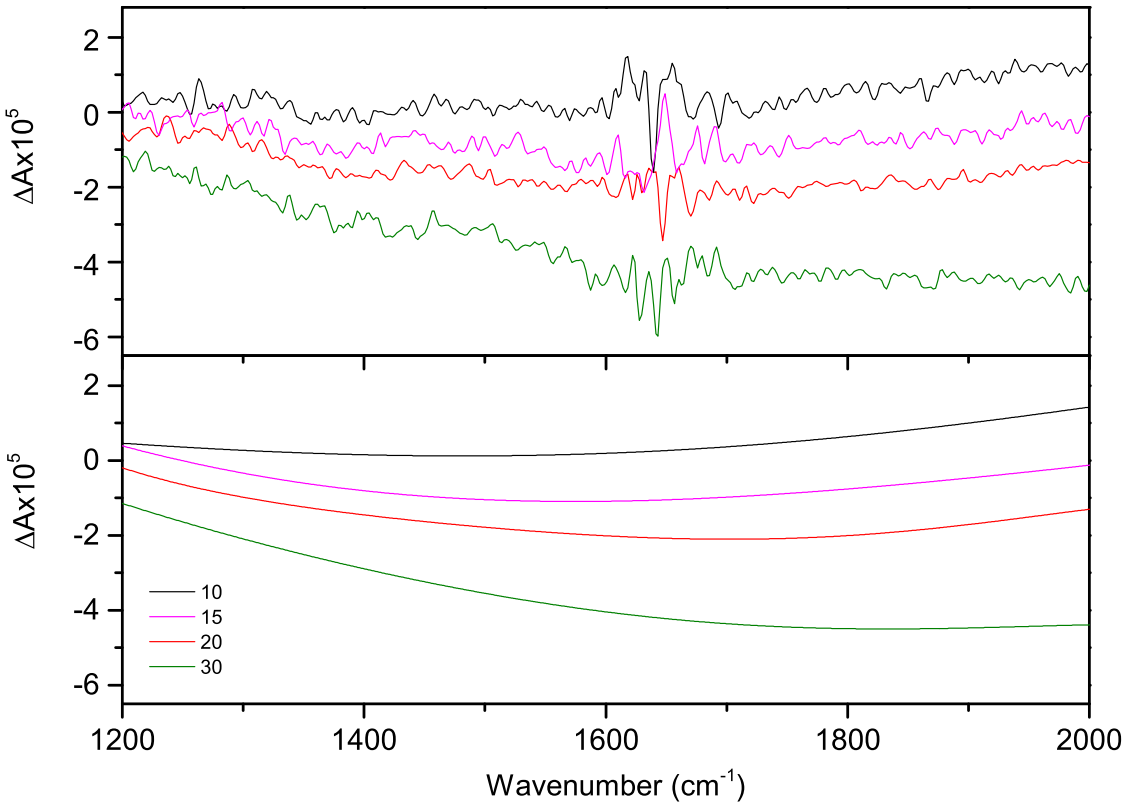


Fig. 6.3: Measured and fitted VCD spectra for different orientations of the top side of the cuvette around value 20° (the numbers stand for rotation of the top glass of the cuvette in degrees).

Results and discussion

As stated earlier and by analyzing Figure 6.2 it can be seen, that the straightest VCD spectrum is around 20° and 100°. By observing the more detailed spectra in Figure 6.3 it was concluded that the ideal position lays between 10-15°. A permanent mark was done on the top glass of the cuvette between orientations 10-15° and in all subsequent measurements the cuvette was assembled in this configuration.

By further examination it has been proven, that by rotating the cuvette by approx. 90° same spectrum can be obtained. In order to prove this statement the values of absorbance at 1800 cm^{-1} were recorded for different orientations of the top glass of the cuvette. The dependence of absorbance at 1800 cm^{-1} on the position of the top glass of the cuvette is plotted in Figure 6.4. The data were fitted by the sine function. The period of the function was determined from the parameters of the fit. The period $T = (94.8 \pm 3.6)°$. Thus it has been proven that the cuvette really behaves in a symmetrical fashion.

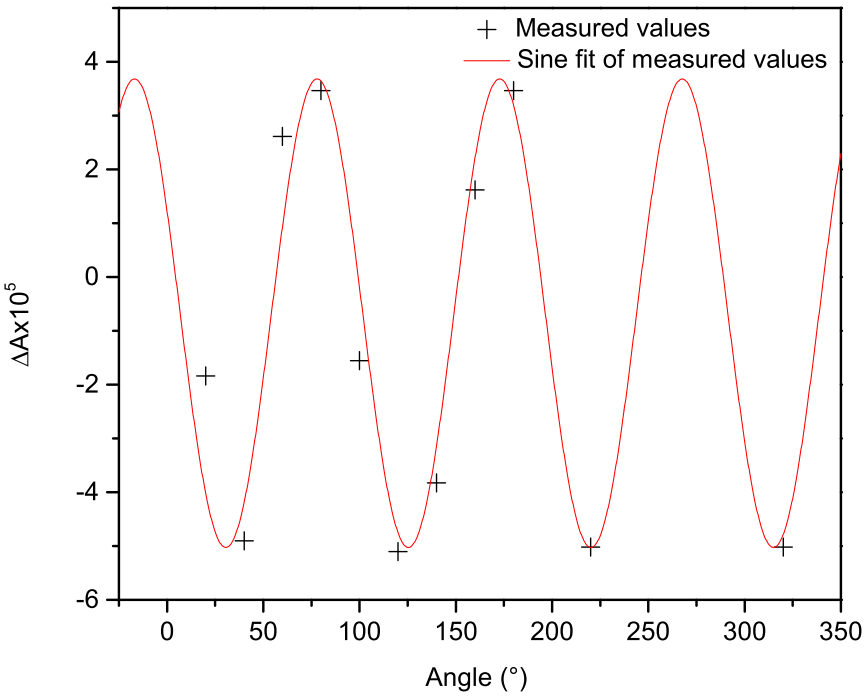


Fig. 6.4: The dependence of absorbance at 1800 cm^{-1} on the position of the top glass of the cuvette. The measured values were fitted by a sine function.

6.2 Solvent dependency study of D-DMT and L-DMT by VA and VCD

In this section VA and VCD spectra of D-DMT and L-DMT in different solvents will be presented. Most importantly measurements in H_2O will be shown, which will be followed by a detailed interpretation by *ab intio* calculations in Section 6.5. Moreover, measurements in D_2O, DMSO and $CHCl_3$ will be presented. The presented spectra will be followed by a detailed comparison of D-DMT in D_2O and H_2O and a comparison between D-DMT in H_2O, DMSO and $CHCl_3$.

The reason why measurements in different solvents were made is to examine the changes in the VA and VCD spectra of D-DMT and L-DMT. Firstly, we investigated the VA/VCD spectra of D-DMT in H_2O and D_2O. Secondly, DMSO and $CHCl_3$ solvents were selected with a purpose. It is known that H_2O is a polar solvent and that DMSO is a less polar solvent and the least polar solvent is $CHCl_3$. We wanted to observe the changes in spectra when using less polar solvents. We also wanted to investigate DMT in CCl_4, however, due to bad solubility of the studied molecule these measurements were not possible.

The main purpose why solvents of different polarity were used is to investigate the presence of hydrogen bondings that seem to be crucial in determination of the conformation of the studied molecule. The molecule may create intra-molecular hydrogen bondings between the O atom of the carbonyl group and the hydrogen atom of the OH group. We assume that this type of bonding will be present in less polar solvents such as $CHCl_3$ and DMSO. On the other hand hydrogen bonding may occur between the hydrogen of the OH group of the studied molecule and O atom of the solvent. We assume that this sort of hydrogen bonding will be present in polar solvent such as water. In this section we wanted to investigate this topic from an experimental side of view.

6.2.1 DMT in H_2O and D_2O

Normalized VA/VCD spectra of D-DMT, L-DMT and LoA spectra in H_2O and D_2O are shown in Figure 6.5. It can be seen that the spectra of D-DMT and L-DMT are almost the mirror images of each other. In case of H_2O solution artifact is present at 1170 cm^{-1}. Same artifact is present in DMSO solution, not present in D_2O and $CHCl_3$. In both solutions small artifact in the C=O stretching region is present.

6.2.2 DMT in DMSO and $CHCl_3$

Normalized VA/VCD spectra of D-DMT, L-DMT and LoA spectra in DMSO and $CHCl_3$ are shown in Figure 6.6. It can be seen that the spectra of D-DMT and L-DMT are almost the mirror images of each other. In case of DMSO small artifact is seen at 1170 cm^{-1}. In case of $CHCl_3$ no substantial artifact is observed.

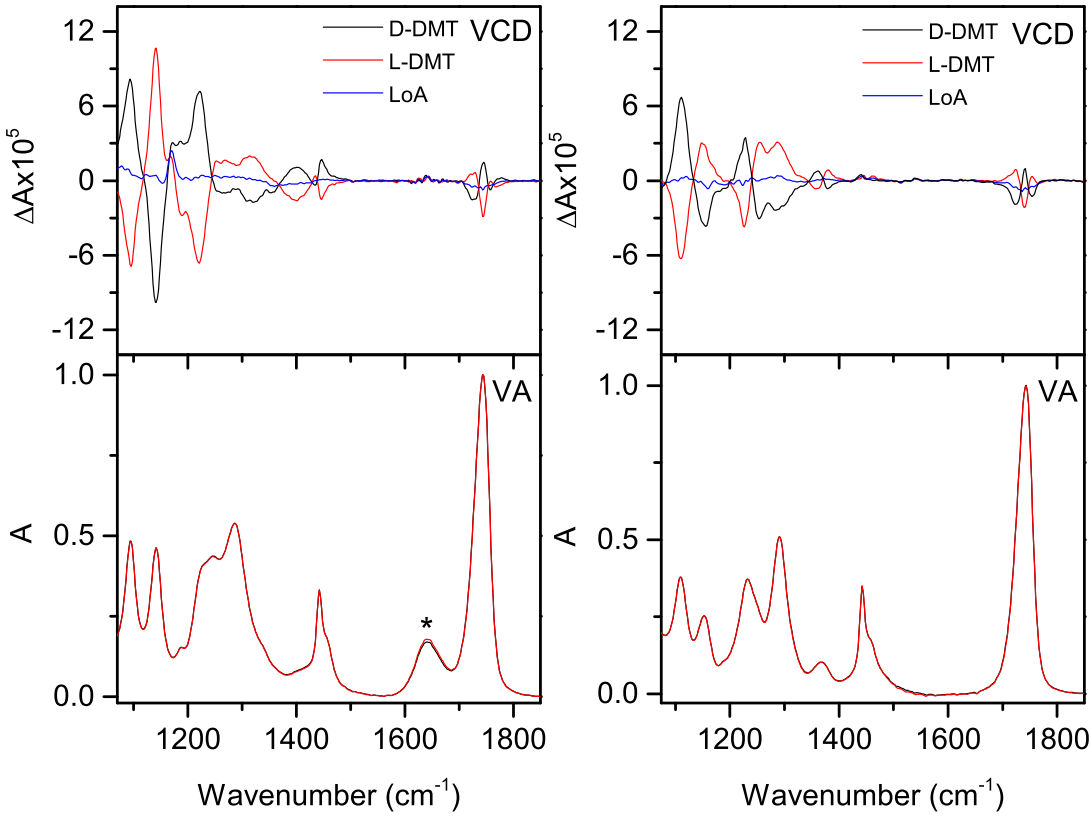


Fig. 6.5: Normalized VCD, VA and LoA spectra of D-DMT and L-DMT in H_2O (left) and D_2O (right). Starred band refers to residual solvent signal.

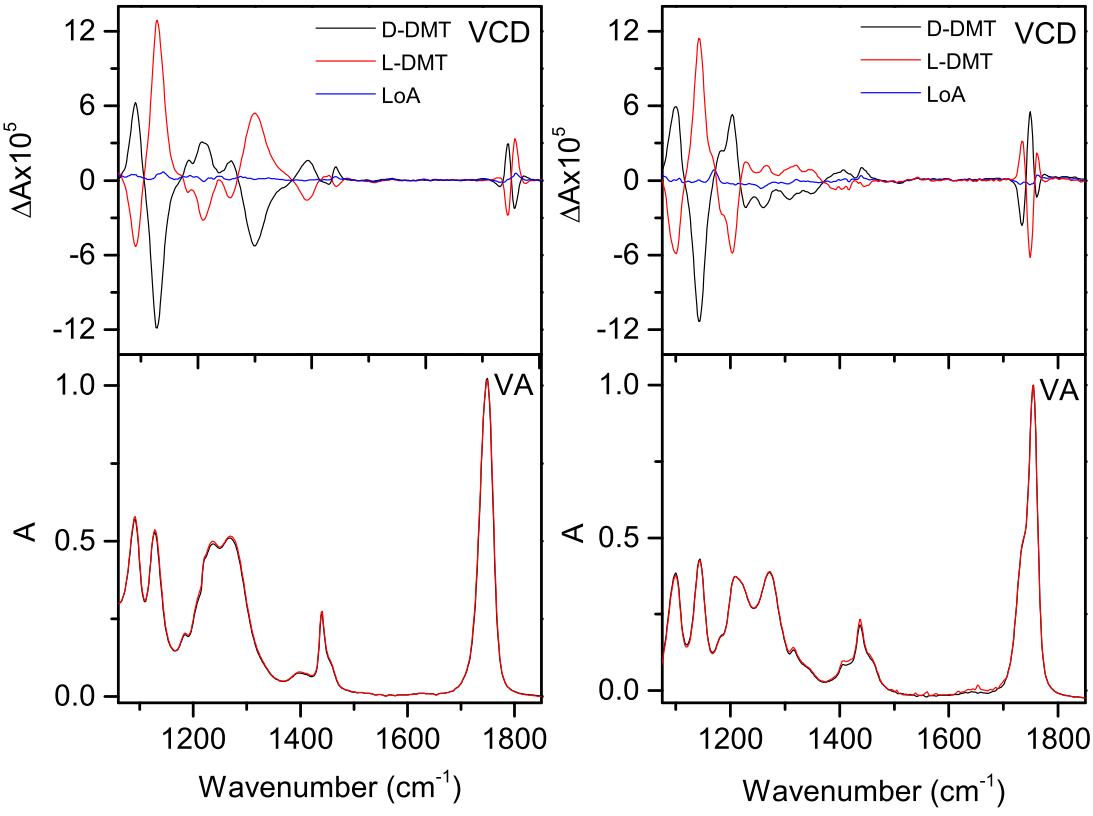


Fig. 6.6: Normalized VCD, VA and LoA spectra of D-DMT and L-DMT in $CHCl_3$ (left) and DMSO (right).

6.2.3 Comparison of VA and VCD spectra of D-DMT in H_2O and D_2O

VA/VCD spectra will be compared in the above mentioned solvents in the following spectral regions. The VA/VCD spectra of D-DMT are shown in Figure 6.7. Table 6.1 may help us to discuss the reasons of various spectral shifts. In Table 6.2 labeled bands have their assigned wavenumbers.

Tab. 6.1: Informative table of vibrations associated with D-DMT [37].

Range (cm^{-1})	Vib. type	Intensity	Assignment
1100	bending	medium	C-C-C
1110-1320	streching	medium-strong	O-C (sometimes 2 peaks)
1350-1470	bending	medium	CH_2 and CH_3 deformation
1370-1390	bending	medium	CH_3 deformation
1300-1430	bending	medium	O-H bending (in-plane)
1300-1440	bending	medium	C-O-H bending
1735-1750 (esters)	stretching	strong	C=O

1. **Spectral region 1600 - 1800 cm^{-1}**

   By comparing **VA** spectrum of H_2O and D_2O we see, that in the spectral region 1600-1800 cm^{-1} very little shifts of vibrational bands are seen. For the band 1c, which corresponds to the C=O stretching vibration, a shift of 1 cm^{-1} to lower wavenumbers is observed in D_2O in comparison with H_2O solution. This effect might be caused by hydrogen bondings between the oxygen atom of C=O group and the hydrogen/deuterium of the solvent. Since the hydrogen bond with deuterium is slightly stronger [38], the frequency of the C=O vibration should decrease, which was experimentally observed. This fact is a very good indicator that hydrogen bonding might be present between the studied molecule and H_2O resp. D_2O. Unlike in H_2O solution, no residual solvent signal is present in D_2O solution.

   In the corresponding **VCD** spectrum in the C=O stretching region even a bigger shift to lower frequencies in D_2O solution is observed (for band 1d a shift of 5 cm^{-1}). In general the spectral profile in D_2O is very similar to the spectral profile in H_2O solution.

2. **Spectral region 1070 - 1600 cm^{-1}**

   First, by examining the **VA** spectrum from higher to lower frequencies it can be seen, that almost no shifts, nor intensity changes are observed in the 1400-1600 cm^{-1} region.

   In the lower spectral region shifts of wavebands is observed. Usually by changing the solvent from H_2O to D_2O a replacement of hydrogen atoms of the hydroxyl group of the molecule by deuterium atoms of the solvent may be expected. As mentioned earlier, the hydrogen bond gets slightly stronger. These changes in D_2O solution lead to an increase in wavenumbers

in vibrational bands. More precisely, an increase in frequencies of several vibrational bands in the region 1070-1400 cm^{-1} is observed. For example, increase in frequencies was observed at bands 5, 6, 6b, 7, 9 and 10.

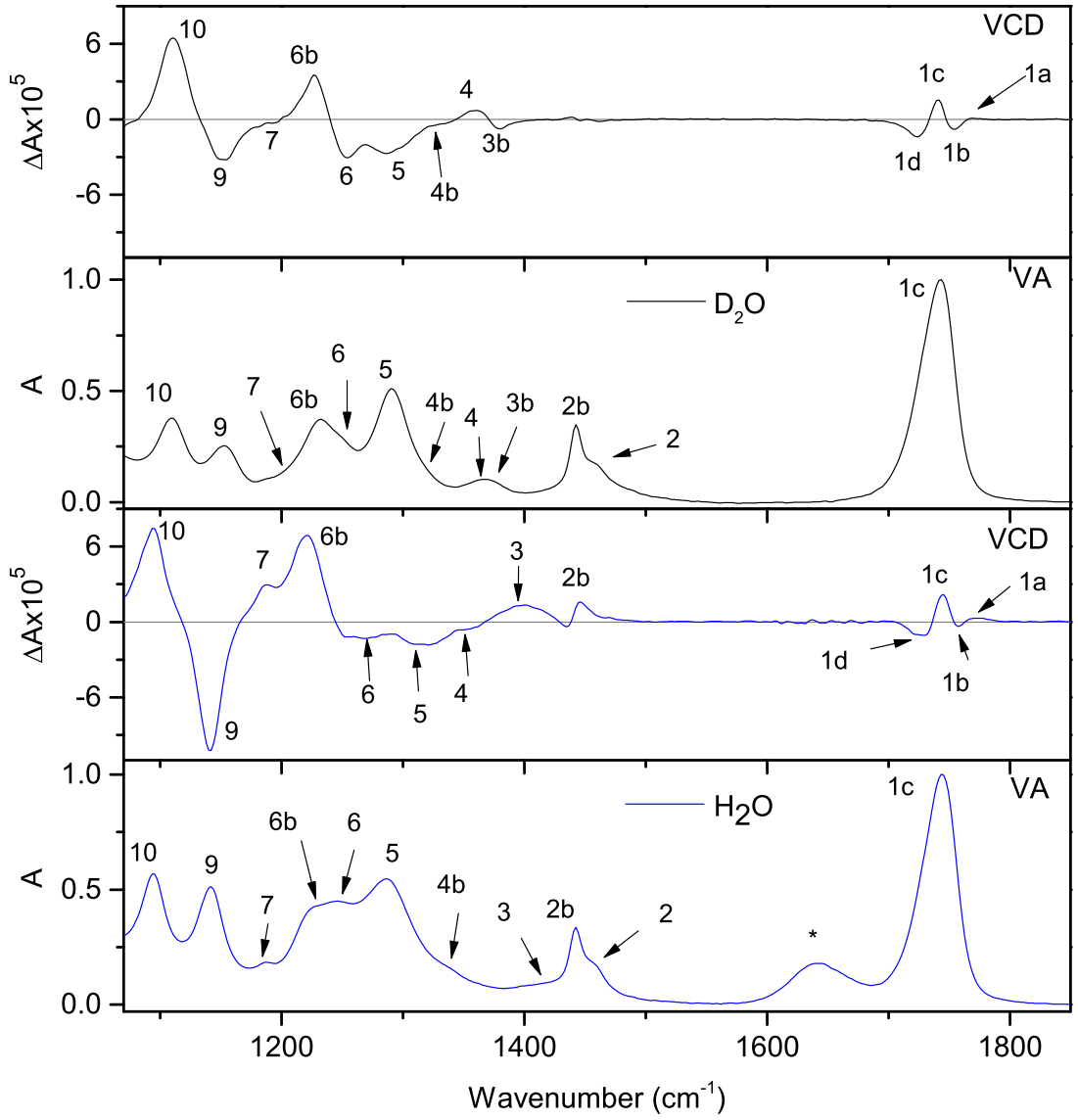


Fig. 6.7: Labeled VA/VCD spectra of D-DMT measured in H_2O (blue spectra) and D_2O (black spectra). Starred band refers to residual solvent signal.

Substantial changes are observed in the corresponding **VCD** spectra. In the higher frequency region there are bands 2b and 3 visible in H_2O but not visible in D_2O. Also the bands 3b and 4 visible in D_2O are not visible in the corresponding H_2O spectrum. In the region 1070-1300 cm^{-1} the bands in D_2O solution are shifted upwards in comparison with H_2O solution and intensity changes are also observed. It is interesting that the band 7 changes its sign by changing the solvent.

6.2.4 Conclusion

Overall the character of DMT spectra remain similar in H_2O and D_2O and the change of solvent does not change substantially the conformation state of the molecule. Nevertheless, larger changes of VCD pattern caused by change of solvent demonstrate enormous sensitivity of VCD to such low perturbation as isotopic exchange.

6.2.5 Comparison of VA and VCD spectra of DMT in H_2O, DMSO and $CHCl_3$

Compared VCD and VA spectra of D-DMT in H_2O, DMSO and $CHCl_3$ are seen in Figure 6.10. In order to better examine the differences in spectra, overlaid VCD and VA in different solvents were plotted and can be seen in Figure 6.9. Similarly as in the above discussed section Tables 6.1 might help with identifying the character of the vibration and in Table 6.2 labeled bands are assigned to their corresponding wavenumbers.

In general, the biggest differences are observed in the C=O stretching region and in the C-O stretching, resp. C-O-H bending region. For better explanation purposes, we created Figure 6.8 that will be commented in the following sections.

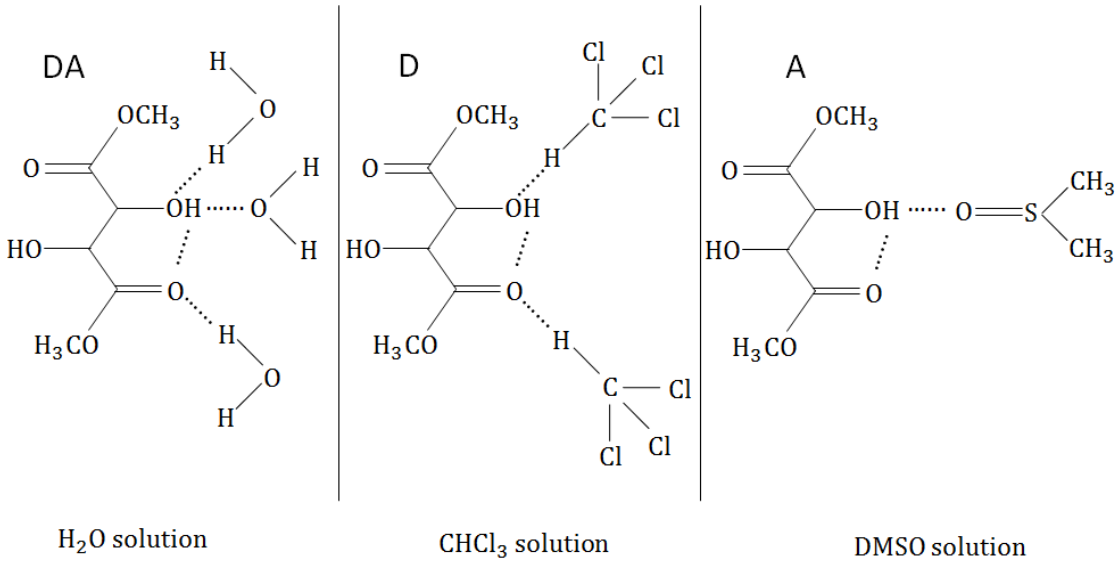


Fig. 6.8: Images of D-DMT and possible interactions with different solvents. D refers to the situation when the solvent molecule is a hydrogen donor, A refers to the situation when the solvent is a hydrogen acceptor and DA refers to the situation when the solvent is a hydrogen donor and acceptor. Dotted lines represent hydrogen bondings.

1. **Spectral region 1600 - 1800 cm^{-1}**

   Major band shifts are seen in the **VA** spectrum in the C=O stretching region when using different solvents. The lowest wavenumber of the C=O stretching is observed in H_2O solution. This may be due to solvent effects since it is known that hydrogen bonds lower the frequency of the C=O stretching vibration. In addition, the shorter the length r of the hydrogen bond, the strongest the H bond link and the lower is the frequency of the

vibration [39]. Since the molecules of water are small, they may be able to get close to the studied molecule and hydrogen bonds between the hydrogens of the solvent and oxygen atoms of the C=O vibration are more preferred than in other solvents.

The band corresponding to the C=O stretching vibration in $CHCl_3$ is shifted to higher frequencies by 5 cm^{-1} in comparison with H_2O solution. According to [40], $CHCl_3$ is able to create hydrogen bonding with oxygen atom. Since $CHCl_3$ is a much bigger molecule in comparison with H_2O, it might be located further from D-DMT. That could be the reason why the hydrogen bond is weaker and a shift to higher frequencies is observed. In Figure 6.8 the hydrogen bonding between the solvent and D-DMT are shown.

In case of DMSO two vibrational bands are observed in the C=O stretching region. The most intensive one has the highest wavenumber among all three solutions. In Figure 6.8 we observe, that DMSO does not create hydrogen bonds with the O atom of the C=O stretching vibration. Thus, that could be the reason why DMSO has the highest wavenumber associated with the C=O stretching vibration.

In addition, in DMSO a less intensive band 1d (see Figure 6.10) at 1734 cm^{-1} appears. According to [9] this could be a contribution from a D-DMT conformer in which intra-molecular hydrogen bonding takes place. Most likely there will be a competition for the hydrogen atom of the hydroxyl group between the oxygen atom of DMSO and of the C=O group.

In the corresponding **VCD** spectrum in the C=O stretching region bigger changes are seen. By observing the VCD spectra from lower to higher wavenumbers it can be seen, that in H_2O and DMSO an intensive negative band is present whereas in $CHCl_3$ a very weak negative band is present. Next, by moving to higher frequencies, in all three solutions a positive and a negative VCD signal is present. Last, at highest wavenumbers a very weak positive band is observed in all three solutions.

2. **Spectral region 1070-1600 cm^{-1}**

   If we approach the **VA** spectra from lower to higher wavenumbers, first two vibrational bands in the region 1070–1150 are distinguishable in every solution. According to Table 6.1 these vibrations may be associated with the C-C-C bending and the C-O stretching vibration. Spectral shifts are observed in this region. It is observed that DMSO bands are shifted to the higher wavenumbers in comparison with $CHCl_3$ and H_2O. Because DMSO cannot form hydrogen bonds with the oxygen atom of the hydroxyl group, the vibration has the highest frequency. Since H_2O and $CHCl_3$ are able to create hydrogen bondings with the oxygen atom, the frequency of the vibration is lower. This might be the possible explanation of spectral shifts in different solutions.

   In the 1150-1350 cm^{-1} region bigger differences in VA spectrum in different solvents take place. In this region very broad spectral bands are observed and they are very poorly distinguishable from each other. However, band

number 5 (see Figure 6.10) can be recognized in all three solutions. Based on the Table 6.1 this band may be associated with the C-O-H resp. O-H bending vibrations. The lowest frequency of this band is observed in $CHCl_3$ solution, higher frequency is in DMSO solution observed and the highest frequency is in H_2O solution observed. According to [41] an increase in frequency of a bending vibration can be observed when it is hydrogen bonded. Based on Figure 6.8, the hydrogen atom of the hydroxyl group might be hydrogen bonded in DMSO and H_2O. Due to this fact the band is shifted to higher frequency in comparison with the non hydrogen bonded H atom of the hydroxyl group in $CHCl_3$.

In region 1350-1600 cm^{-1} CH_2, CH_3 bending vibrations are present. Small effect is observed on the vibrational bands in this region when changing the solvent.

In general, the **VCD** spectra are very similar to each other in H_2O and DMSO. Substantial differences are observed in the $CHCl_3$ solution, where even changes of the sign of band number 5 is observed and a very intensive band labeled 4b is observed. This major change in the 1300 cm^{-1} region might be caused by different character of the solvent. This region might be associated with the C-O-H, resp. O-H bending vibration. By examining Figure 6.8 we can see that in $CHCl_3$ solution the hydrogen atom is not bound by the solvent, whereas in DMSO and H_2O it is. Thus this could be the reason of the major change of the VCD spectrum in the $CHCl_3$ solution.

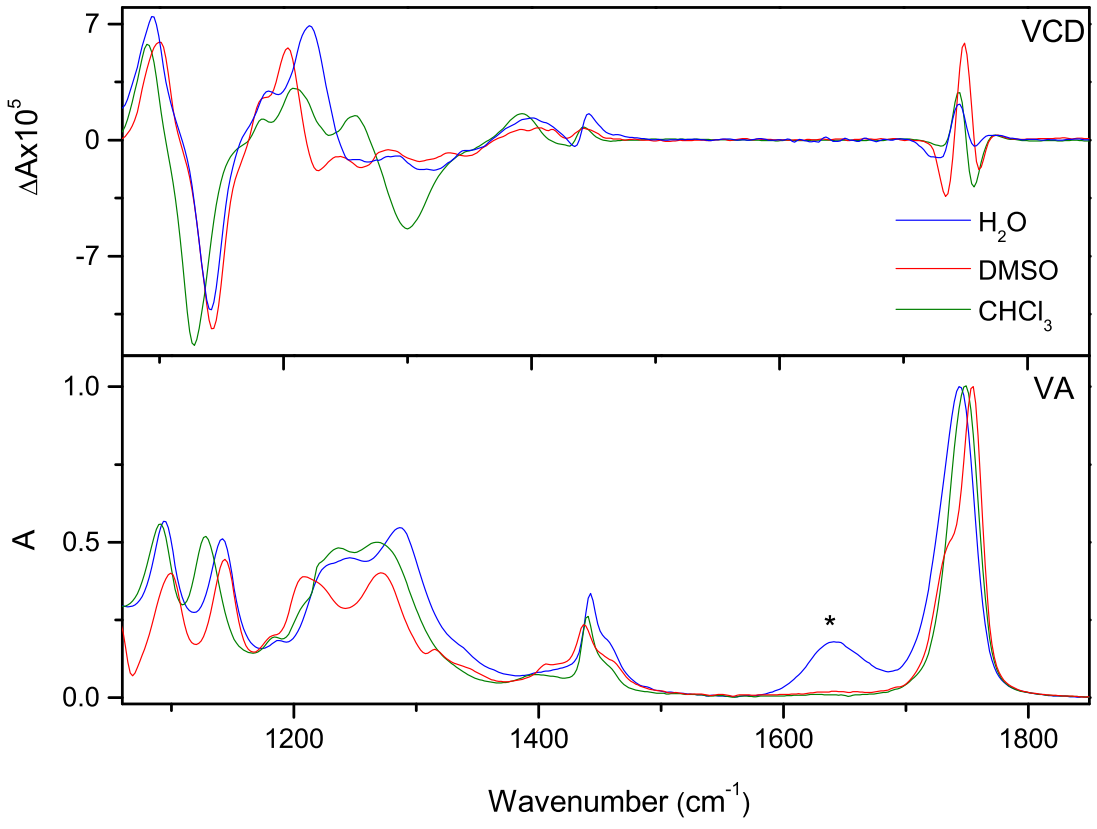


Fig. 6.9: Overlaid VCD and VA of D-DMT in different solvents in H_2O, DMSO and $CHCl_3$. Starred band refers to residual solvent signal.

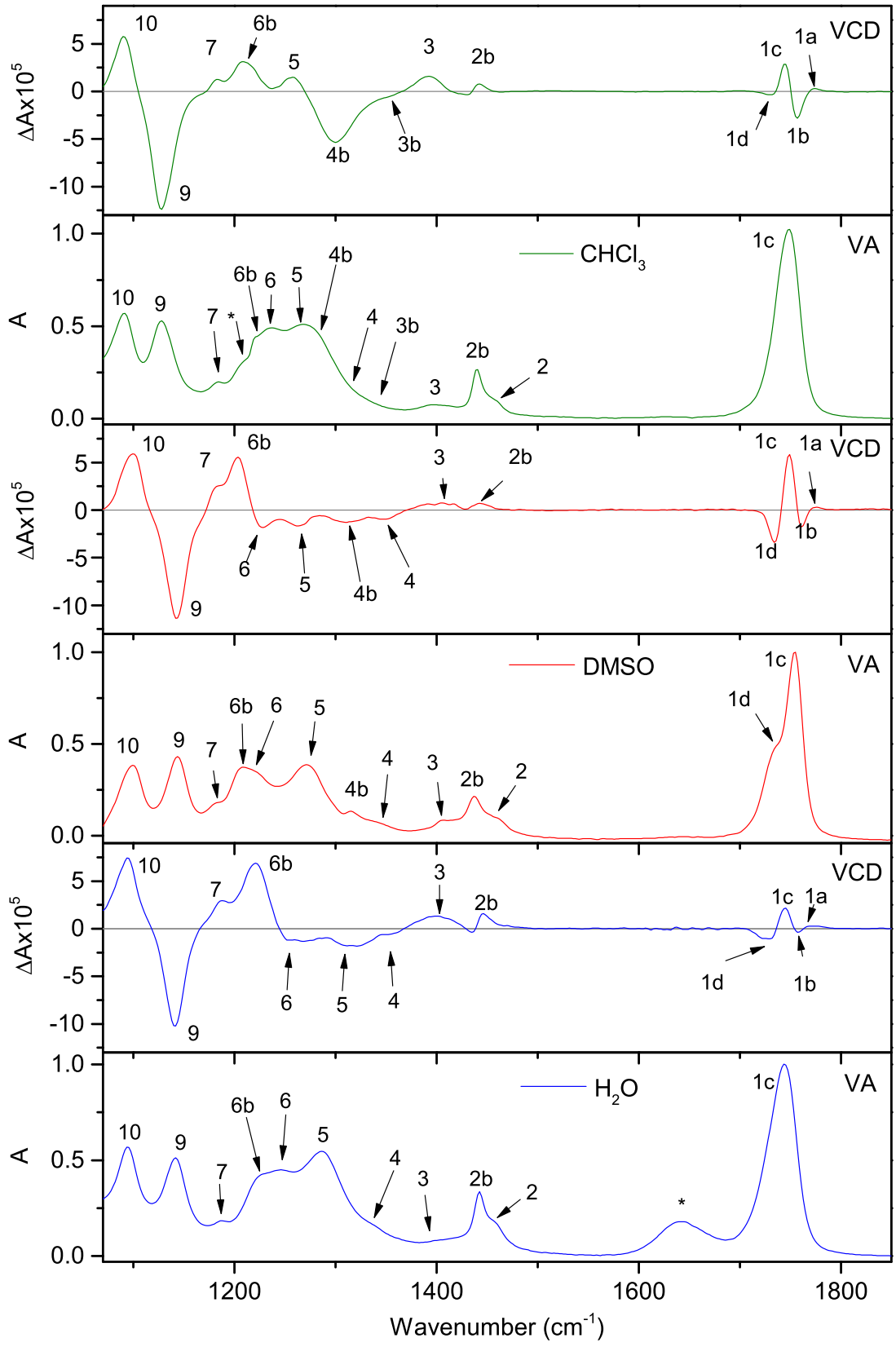


Fig. 6.10: Labeled VA and VCD spectra of D-DMT in different solvents. Starred band refers to residual solvent signal.

Tab. 6.2: Summary of VA and VCD bands of D-DMT in different solvents. Broad ROA bands where no precise wavenumber could be assigned are labeled with a star.

Solvent	H_2O		D_2O		DMSO		$CHCl_3$	
Spectrum	VA	VCD	VA	VCD	VA	VCD	VA	VCD
Unit	(cm^{-1})	(cm^{-1})	(cm^{-1})	(cm^{-1})	(cm^{-1})	(cm^{-1})	(cm^{-1})	(cm^{-1})
1a	-	1768	-	1767	-	1776	-	1774
1b	-	1756	-	1754	-	1760	-	1757
1c	1746	1744	1745	1741	1755	1749	1751	1745
1d	-	1729	-	1724	1731	1733	-	1730
2	1463	-	1462	-	1462	-	1459	-
2b	1443	1447	1443	-	1436	1442	1439	1442
3	1398	1398	-	-	1404	1404*	1394	1392
3b	-	-	1370	1379	-	-	1359	1355*
4	1340	1350*	1369	1361	1343	1347*	1333	-
4b	-	-	1323	1323	1316	1313*	1282	1300
5	1289	1289*	1291	1286	1274	1264*	1266	1257
6	1245	1250*	1249	1253	1223	1226	1234	1218
6b	1220	1221	1231	1226	1204	1203	1222	-
7	1185	1188	1204	1197	1180	1184	1182	1184
9	1142	1142	1153	1153	1143	1142	1128	1128
10	1094	1094	1109	1110	1100	1100	1091	1090

6.2.6 Conclusion

In this section VA and VCD spectra of L-DMT and D-DMT in different solvents were measured. It has been shown that the VA and VCD spectra are solvent dependent. We have discussed the reason of various shifts in the VA/VCD spectra of D-DMT in different solvents. The biggest changes in both VA and VCD spectra are observed in the C=O stretching region and in the C-O stretching, C-O-H bending resp. O-H bending region. The changes might be caused by creating of hydrogen bondings between D-DMT and the solvent. In general these measurements gave us important information of the solvent effects on D-DMT and they demonstrate the enormous sensitivity of VA/VCD on conformational changes on DMT.

6.3 Solvent dependency study of D-DMT and L-DMT by Raman scattering and ROA

The same motivation of measurements of D-DMT and L-DMT applies to measurements by Raman scattering and ROA in different solvents. Again, for us the most important are the measurements in H_2O solution which are going to be interpreted by *Ab initio* calculations in Section 6.5. Moreover, measurements of DMT in DMSO and $CHCl_3$ were carried out. One of the main reasons why Raman/ROA measurements were carried out is that they allow us to examine

vibrations in a much lower wavenumber region in comparison with VA/VCD measurements.

Firstly, water solutions of D-DMT and L-DMT were measured on the ICP-CU. Measurements on this spectrometer were performed in other solvents, however, due to strong solvent signal and poor signal/noise ratio the results were not usable. By changing the grating in the spectrometer measurements of D-DMT in H_2O, $CHCl_3$ and DMSO were attempted in the C-H stretching region, nevertheless, there were reasonable Raman spectra measured only in H_2O solution. In case of $CHCl_3$ and DMSO strong interference between the solvent and DMT signal was present. Due to small signal/noise ratio ROA spectra in the C-H stretching region were not usable.

That is why measurements in different solvents including water were measured on the commercial ROA spectrometer Chiral$RAMAN-2X^{TM}$ (SCP-Biotools) with successful results apart from the C-H stretching region.

6.3.1 Comparison of Raman and ROA spectra of D-DMT measured on the SCP-Biotools and ICP-CU

Since Raman and ROA spectra of DMT in H_2O were measured on two different spectrometers, it is useful to make a comparison in order to see potential differences in the obtained spectra. In Figure 6.11 normalized Raman and ROA spectra of D-DMT measured on two different spectrometers are shown.

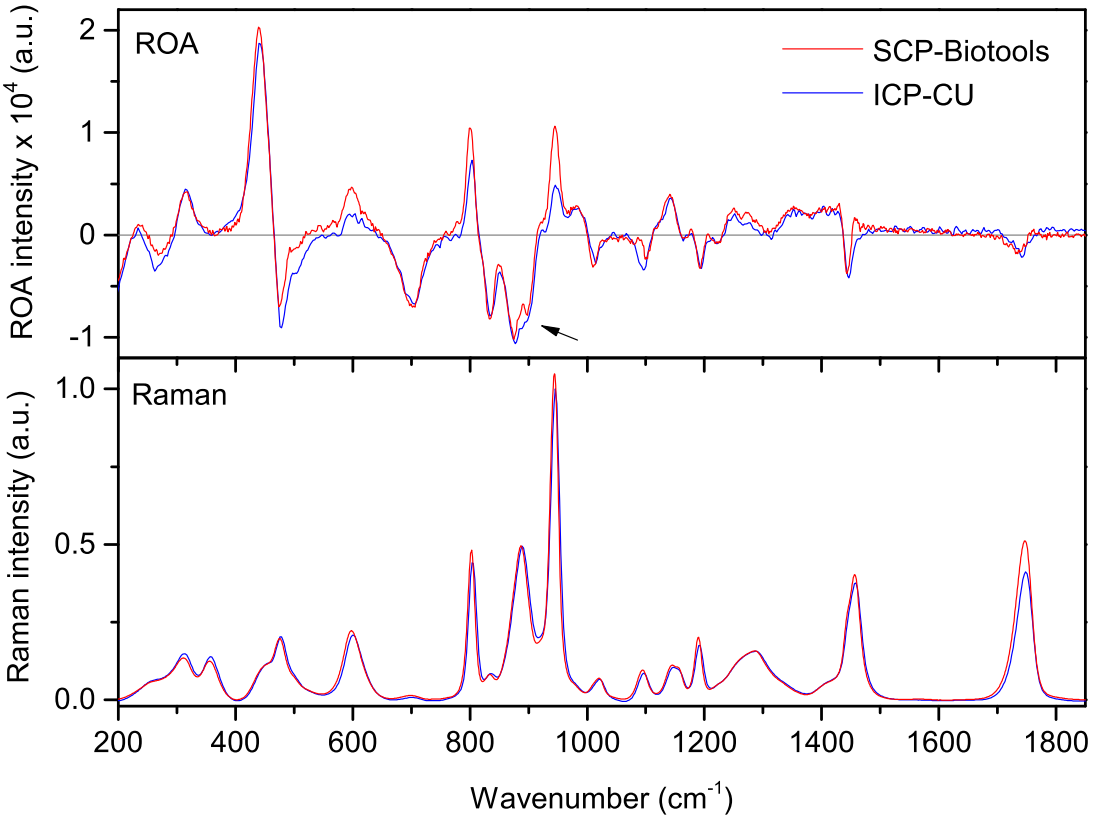


Fig. 6.11: Normalized ROA and Raman spectra of D-DMT in H_2O measured on the commercial SCP-Biotools and on the ICP-CU. The arrow demonstrates slightly better spectral resolution on ICP-CU.

Due to different half-width of the measured spectra on different instruments, the spectra were normalized by their integral intensities. Very good agreement of the Raman spectra is observed. In the corresponding ROA spectra the positions of each bands are almost the same with minor intensity changes. On the ICP-CU slightly better spectral resolution is obtained (for example see the region with the arrow, where on the home-built instrument clearly two distinguishable bands are observed in comparison with the one broad band from the commercial instrument) but at longer accumulation time in comparison with the commercial spectrometer. This statement is in agreement with the measurements in work [42].

Since successful measurements in other solvents were carried out on SCP-Biotools, in the following sections only results obtained from this spectrometer will be presented.

6.3.2 DMT in H_2O

Normalized Raman/ROA spectra of D-DMT, L-DMT and LoA spectra in H_2O are shown in Figure 6.12. It can be seen that the D-DMT spectrum is almost the mirror image of the L-DMT spectrum. However, several artifacts can be seen in the spectra. Three artifacts in 780-980 cm^{-1} region and one at 1750 cm^{-1} are visible. Artifacts at similar position occur in $CHCl_3$ solution, however, they are not visible in DMSO solution. ROA spectra of DMT in water were reported in [26], nevertheless, in our work much better spectral resolution and substantial increase in signal/noise was reached.

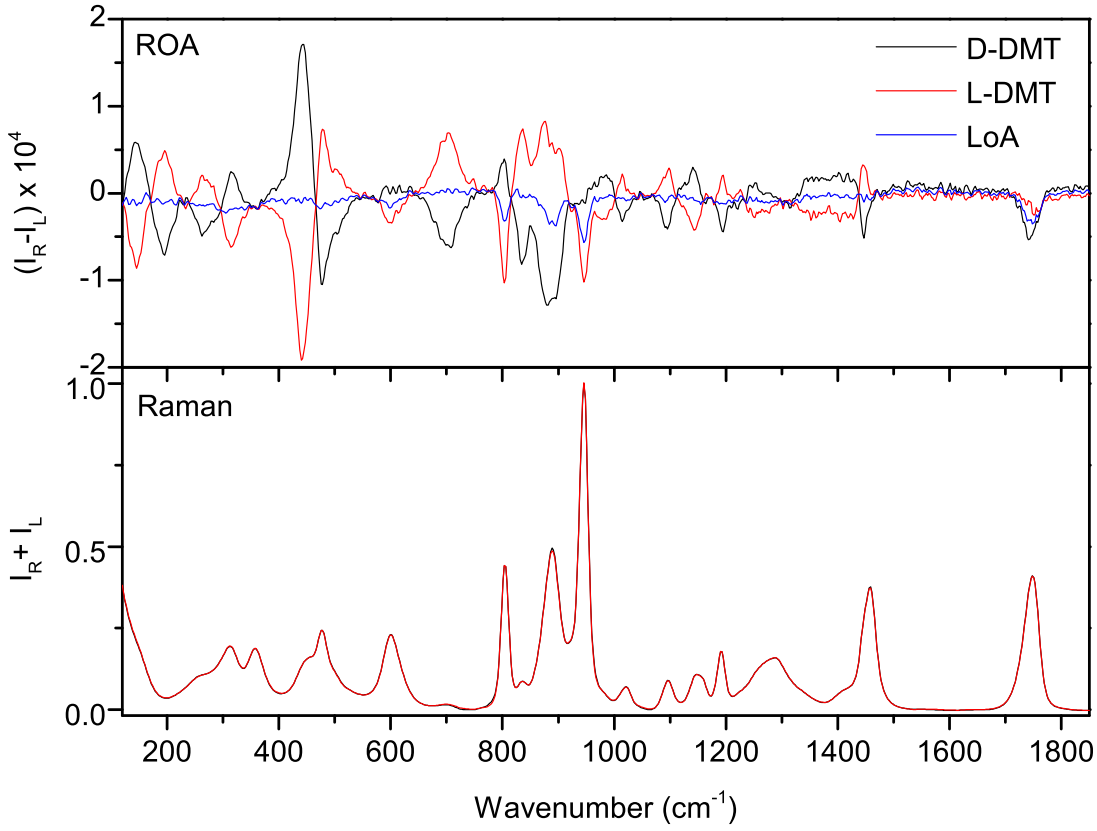


Fig. 6.12: Normalized Raman, ROA and LoA spectra of D-DMT and L-DMT in H_2O.

6.3.3 DMT in DMSO

To our knowledge no measurements on the studied molecules were carried out in this solvent. Normalized Raman/ROA spectra of D-DMT, L-DMT and LoA spectra in DMSO are shown in Figure 6.13. No substantial artifact is present.

Organic solvents (such as $CHCl_3$ or DMSO) are known for their intensive bands in Raman spectra. The detector was saturated in the regions where solvent signal was present in order to obtain good quality spectra of the studied molecule which has much weaker signal than the solvent. That is the reason why the saturated regions are left out from the spectra.

The most interesting information from this spectrum is the absence of artifact at ~1750 cm^{-1}. This information is valuable because it is an indicator that real ROA signal of the studied molecule is visible at this wavenumber. Mirror symmetry in the region is observed, whereas no mirror symmetry was observed in the other solutions.

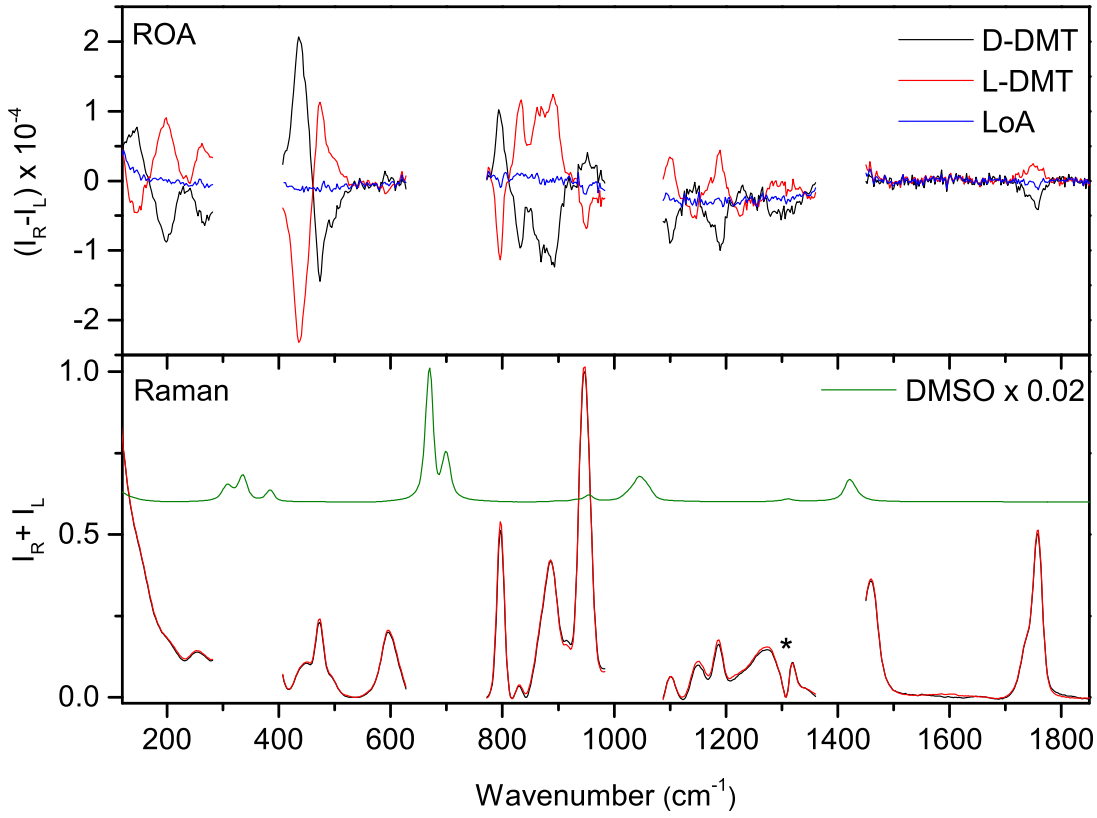


Fig. 6.13: Normalized Raman, ROA and LoA spectra of D-DMT and L-DMT in DMSO including DMSO solvent spectrum. Starred region refers to residual solvent signal.

6.3.4 DMT in $CHCl_3$

To our knowledge no Raman nor ROA spectra were reported in $CHCl_3$. Normalized Raman/ROA spectra of D-DMT, L-DMT and LoA spectra in

$CHCl_3$ are shown in Figure 6.14. Small artifacts at similar positions are visible in H_2O solution.

Similarly as in DMSO the detector was saturated in the regions where solvent signal was present. That is the reason why no Raman/ROA signal is available in this region.

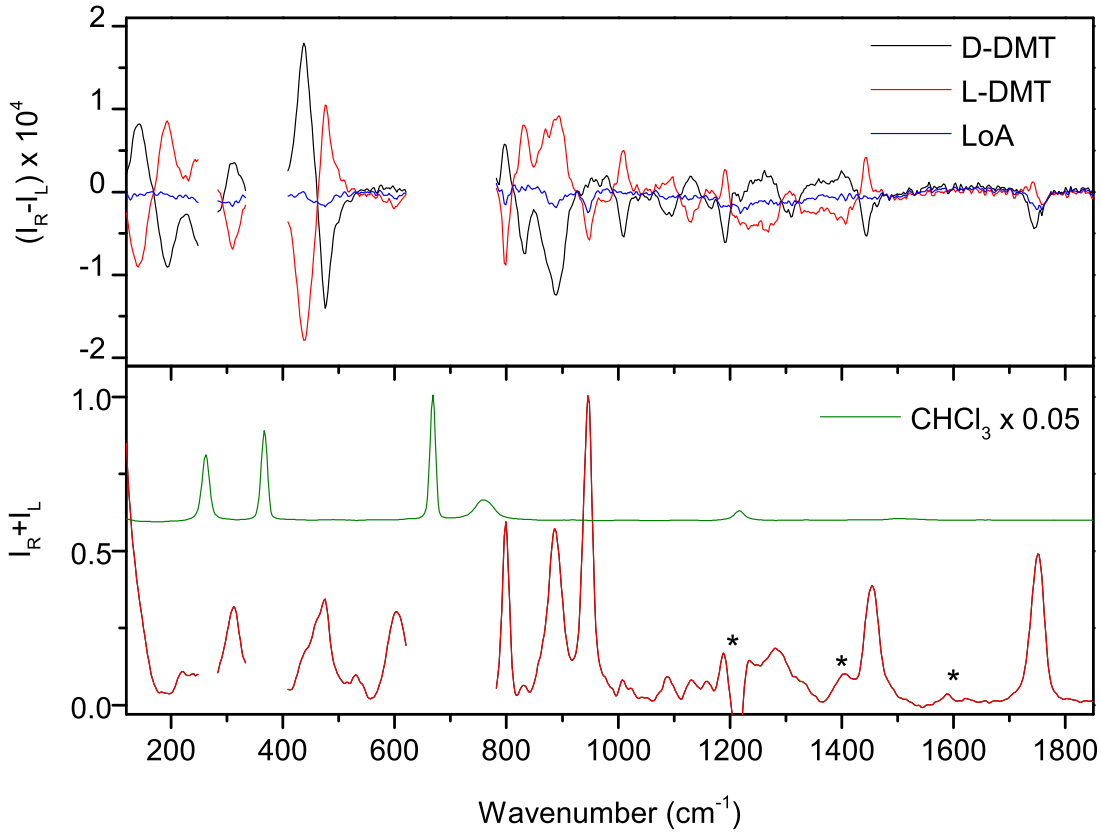


Fig. 6.14: Normalized Raman, ROA and LoA spectra of D-DMT and L-DMT in $CHCl_3$ including $CHCl_3$ solvent spectrum. Starred regions refer to residual solvent signal.

6.3.5 Comparison of ROA spectra of D-DMT in H_2O, DMSO and $CHCl_3$.

In Figure 6.16 compared and labeled Raman/ROA spectra of D-DMT in H_2O, $CHCl_3$ and DMSO can be seen. In Table 6.3 labeled bands are assigned to their corresponding wavenumbers. In order to compare spectra of the studied molecule in different solvents better, overlaid ROA spectra are shown in Figure 6.15.

In comparison with VA and VCD, Raman scattering and ROA measurements allow us to measure in a much wider spectral region. On the other hand more noise is present in the Raman/ROA spectra which makes the identification of each vibrational bands more difficult. The spectra will be interpreted in the following spectral regions.

1. **Spectral region 1070-1850 cm^{-1}**

   The biggest differences between the spectra of D-DMT in different solvents can be seen in the highlighted spectral regions. These regions correspond to the C=O stretching, C-O stretching, C-O-H bending resp. O-H bending vibrations. In case of the C=O stretching region the profile of the Raman spectrum of all three solutions is almost the same as in case of the corresponding VA spectrum. Nevertheless, only one negative ROA band is observed for all three solutions in comparison with four bands in the corresponding VCD spectra.

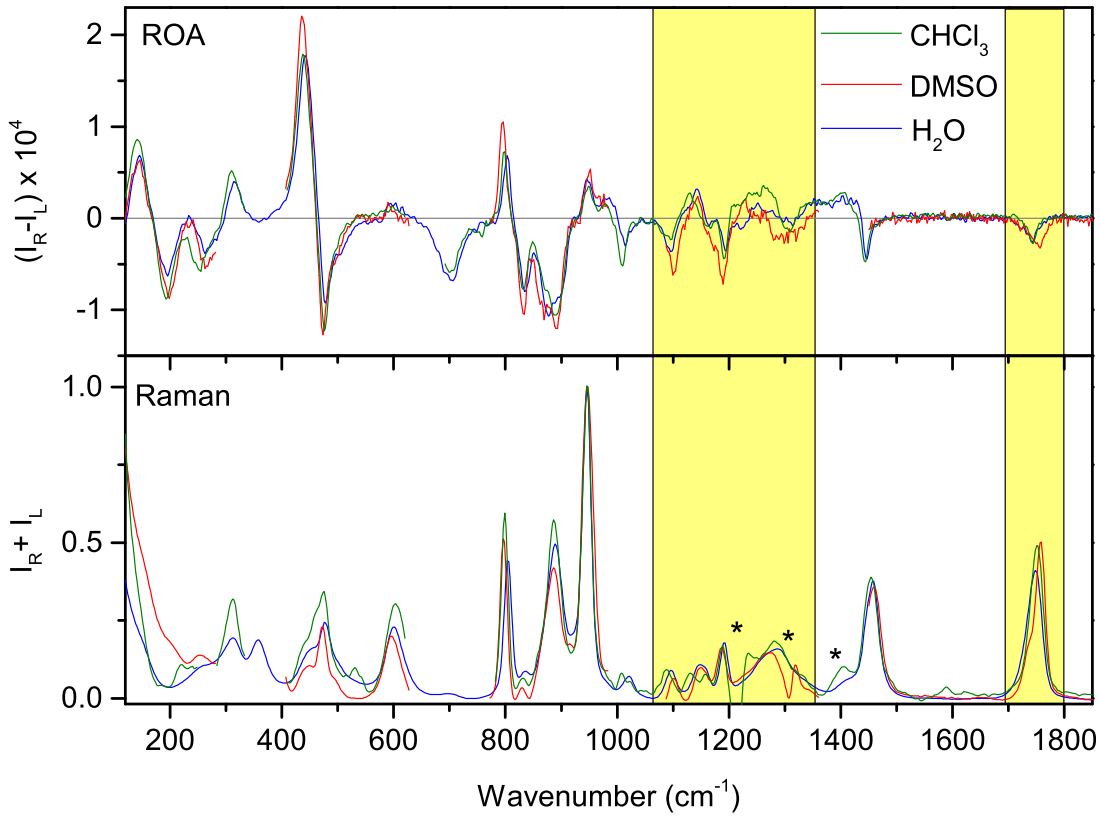


Fig. 6.15: Overlaid spectra of D-DMT in H_2O, DMSO and $CHCl_3$. Starred regions refer to residual solvent signal. Biggest changes of Raman/ROA spectra in different solutions have been highlighted.

   In case of the C-O stretching, C-O-H bending resp. O-H bending vibrations in the 1100-1400 cm^{-1} region, the Raman spectra cannot be interpreted precisely due to strong solvent signal. Significant changes are observed in the corresponding ROA spectrum although the interpretation of band shifts is difficult due to noise in the spectrum.

2. **Spectral region 120-1070 cm^{-1}**

   In case of Raman/ROA measurements the most interesting spectral region is from 120-1070 cm^{-1} where VA/VCD spectrum cannot be measured. In

this spectral region mostly skeletal vibrations are present. Firstly, by observing the spectral profile of all three solutions, no substantial difference is observed. Nevertheless, minor changes by changing the solvent are observed. For example, there is a shoulder at approx. 500 cm^{-1} in H_2O observed (labeled with number 20 in Figure 6.16) as well as in DMSO, while no band in this frequency region is present in $CHCl_3$. In DMSO the best spectral resolution is achieved. This might be caused by the size of the solvent molecule. By that we mean that smaller interaction between the solvent and the molecule takes place, thus the studied molecule stays in a more rigid form. Minor changes in position and intensity of the bands is observed. On the other hand, no particular shifts of band is observed. The changes in the Raman spectra in the 200–700 cm^{-1} are caused mainly by problems with subtraction of much stronger solvent signal.

6.3.6 Conclusion

By comparing the Raman/ROA spectra no particular changes are observed in the 120–1070 cm^{-1} spectral region. Thus different solvents do not effect substantially the skeletal vibrations of D-DMT. The largest changes are present in the 1100–1300 cm^{-1} spectral region with C-O stretching, C-O-H bending vibrations and these changes are also in the C=O stretching region. In general, Raman/ROA measurements allow us to measure wider spectral region, nevertheless, the signal/noise ratio is substantially lower in comparison with VA/VCD measurements.

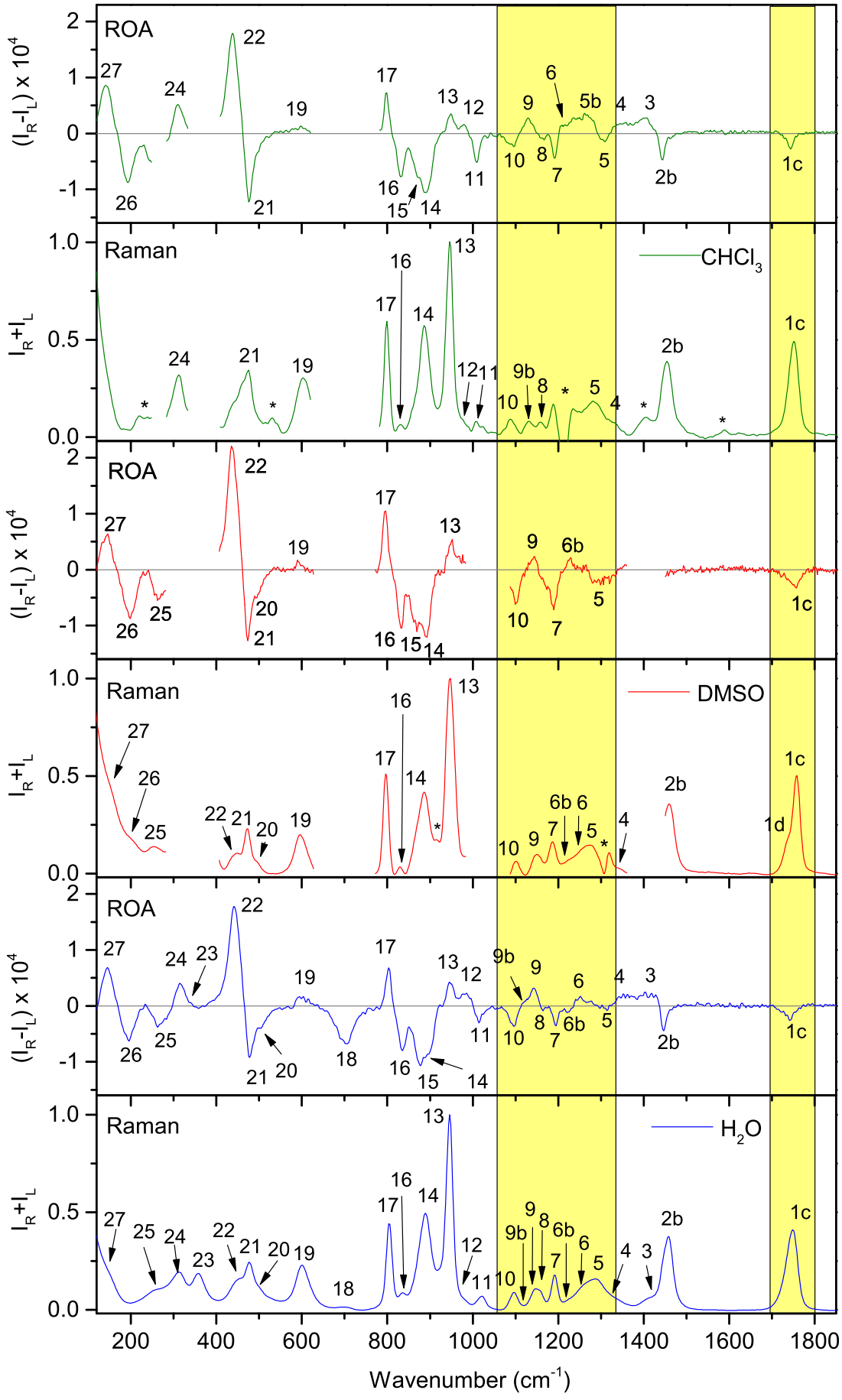


Fig. 6.16: Labeled ROA and Raman spectra of D-DMT in different solvents. Starred regions refer to residual solvent signal. Biggest changes of Raman/ROA spectra in different solutions have been highlighted.

Tab. 6.3: Summary of Raman and ROA bands of D-DMT in different solvents. Broad ROA bands where no precise wavenumber could be assigned are labeled with a star.

Solvent	H_2O		DMSO		$CHCl_3$	
Spectrum Unit	Raman (cm^{-1})	ROA (cm^{-1})	Raman (cm^{-1})	ROA (cm^{-1})	Raman (cm^{-1})	ROA (cm^{-1})
1c	1751	1741	1758	1756	1750	1743
1d	-	-	1732	-	-	-
2b	1458	1446	1462	-	1454	1444
3	1406	1406*	-	-	-	1406*
4	1340	1340*	1345	-	1331	1340*
5	1291	1313	1280	1291*	1282	-
6	1264	1252	1254	1227	-	-
6b	1225	1222	1225	-	-	-
7	1191	1194	1187	1189	-	1191
8	1158	1164	-	-	1158	-
9	1143	1143	1145	1144	-	1131
9b	1120	1120	-	-	1128	-
10	1096	1095	1100	1099	1083	1097
11	1022	1014	-	-	1013	1010
12	984	984*	-	-	985	976
13	946	946	946	952	945	948
14	889	891	888	893	888	890
15	-	877	-	869	-	875
16	835	835	829	834	833	832
17	803	803	796	796	800	799
18	703	705	-	-	-	-
19	599	599*	594	589	602	596
20	505	504	502	492	-	-
21	478	477	473	473	472	476
22	447	442	442	435	-	438
23	357	342	-	-	-	-
24	314	314	-	-	310	310
25	253	262	251	262	-	-
26	-	196	205	198	-	193
27	152	145	156	146	-	141

6.3.7 Comparison of the Raman/VA and ROA/VCD spectra of D-DMT in H_2O

A comparison of the Raman/VA and ROA/VCD spectra of D-DMT in H_2O is shown in Figure 6.17. By comparing Raman/VA spectra it is observed, that almost every band present in the VA spectrum is seen in the Raman spectrum. Tiny spectral shifts are observed between the spectra but in general very good agreement in band positions is achieved.

Similarities in ROA and VCD spectra of D-DMT in the same Figure are also observed. Generally the VCD spectra have better signal/noise ratio in comparison with the corresponding ROA spectra. On the other hand ROA allows us to observe bands in much broader spectral region.

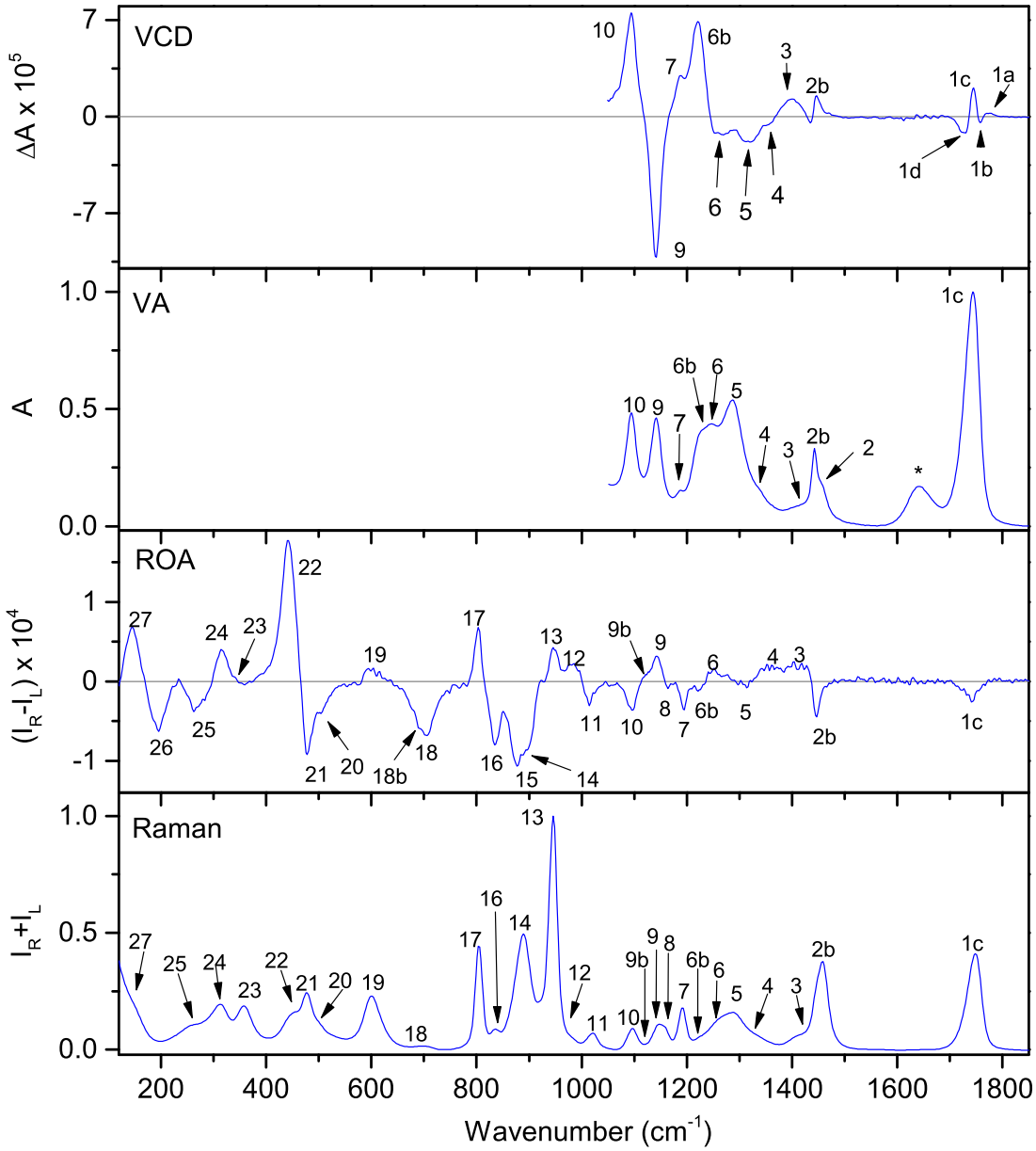


Fig. 6.17: Labeled ROA/Raman and VA/VCD spectra of D-DMT in H_2O. Symbol * refers to residual solvent signal.

6.3.8 DMT in H_2O measured in the C-H stretching region

There were also Raman spectra of D-DMT and L-DMT in the C-H stretching region measured on the ICP-CU. To date no measurements were reported in the C-H stretching region of DMT. In Figure 6.18 the spectrum of D-DMT and L-DMT in the C-H stretching region is shown. Four major vibrational bands are observed in the spectrum. As expected, the Raman spectra of both enantiomers are the same.

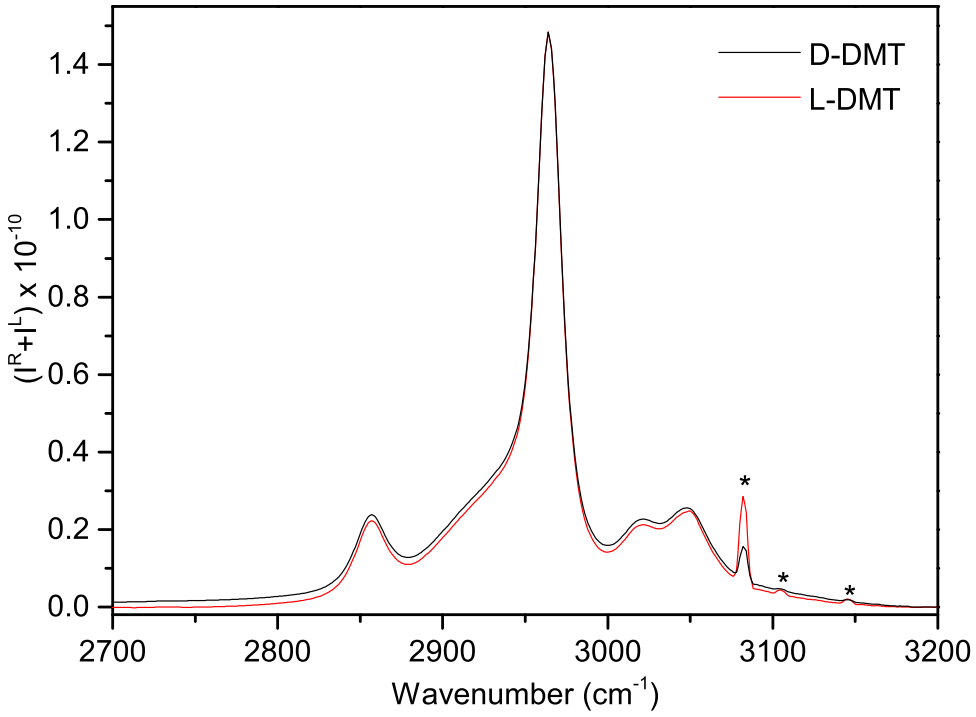


Fig. 6.18: D-DMT and L-DMT in H_2O in the CH stretching region. Starred bands correspond to plasma lines of argon laser.

We tried to investigate the Raman/ROA when dissolving DMT in $CHCl_3$ and DMSO. There was a very strong solvent signal present in this region which strongly interfered with DMT signal. ROA spectra are not presented here since poor signal/noise ratio was achieved.

6.4 *Ab initio* calculations of D-DMT in H_2O

In this section a step by step process of the calculation of the VA/VCD and Raman/ROA spectra of D-DMT will be introduced. The conformation analysis was carried out on D-DMT enantiomer. The calculations were performed by the *Gaussian 09* software [13].

6.4.1 Conformation analysis

Since D-DMT is a very flexible molecule, more possible conformation may coexist in aqueous solution. Our goal was to find all the possible conformations that could be present in the solution. During the conformation analysis the dihedral angles containing the hydroxyl group were fixed, as shown in Figure 6.20 (circled dihedral angle). The reason why we have done this is to prevent formation of intra-molecular H-bonding which would lead to distortion of resulting potential energy scan profiles. Conformation analysis was carried out using the B3LYP functional, 6-31G basis set and CPCM water model.

Firstly, dihedral angle α was rotated from -180°–180° with a step of 10° and in each step geometry optimization was performed.

Two minima were found when rotating dihedral angle α, one at 10° (labeled c) and one at 180° (labeled t), see Figure 6.21, subfigure a). Labeling of each dihedral angle value was done according to Figure 6.19. Since the energy of the molecule was much higher when dihedral angle α = c (more than 6 kcal/mol in comparison with dihedral angle t), we assumed, that this dihedral angle will not be favored by the molecule and hence it was not considered for further calculations. The scan of dihedral angle ε is not shown since it behaves in the same fashion as dihedral angle α and similarly only dihedral angle ε = t was considered for further calculations.

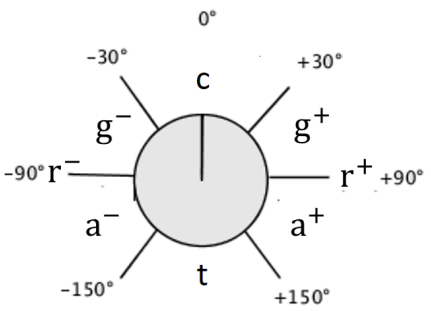


Fig. 6.19: IUPAC notation for different dihedral angles [43].

The molecule has three more degrees of freedom, i.e. potential energy minima for dihedral angles β, γ and δ are needed to be found. For given dihedral angles $\alpha = \varepsilon$ = t potential energy scan was performed for dihedral angle γ. Three minima were found, at $-60°$ (labeled g^-), at 60° (labeled g^+) and at 180° (labeled t), as shown in Figure 6.21, subfigure b).

For a given dihedral angle γ dihedral angle β was rotated and in each step of the rotation geometry optimization was performed.

Firstly, for $\gamma = g^-$ four potential energy minima were found, at -120° (labeled c^-), at -50° (labeled g^-), at 90° (labeled c^+) and at 180° (labeled t), see Figure 6.21, subfigure c).

Secondly, for dihedral angle $\gamma = g^+$ three minima were found, at $-180°$ (labeled t), at $-80°$ (labeled c^-) and at 50° (labeled g^+), see Figure 6.21, subfigure d).

Thirdly, for dihedral angle γ = t two potential energy minima were found, at

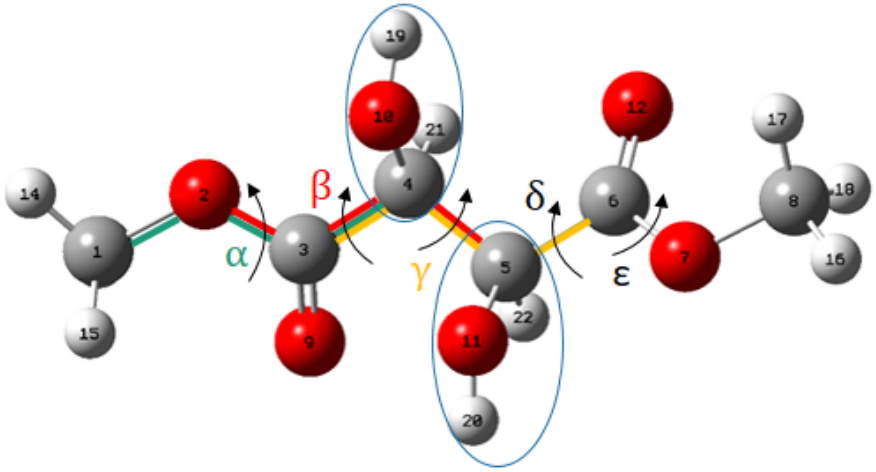


Fig. 6.20: Labeled and colored dihedral angles of D-DMT. Colors of dihedral angles correspond with the potential energy scans shown in the figure below.

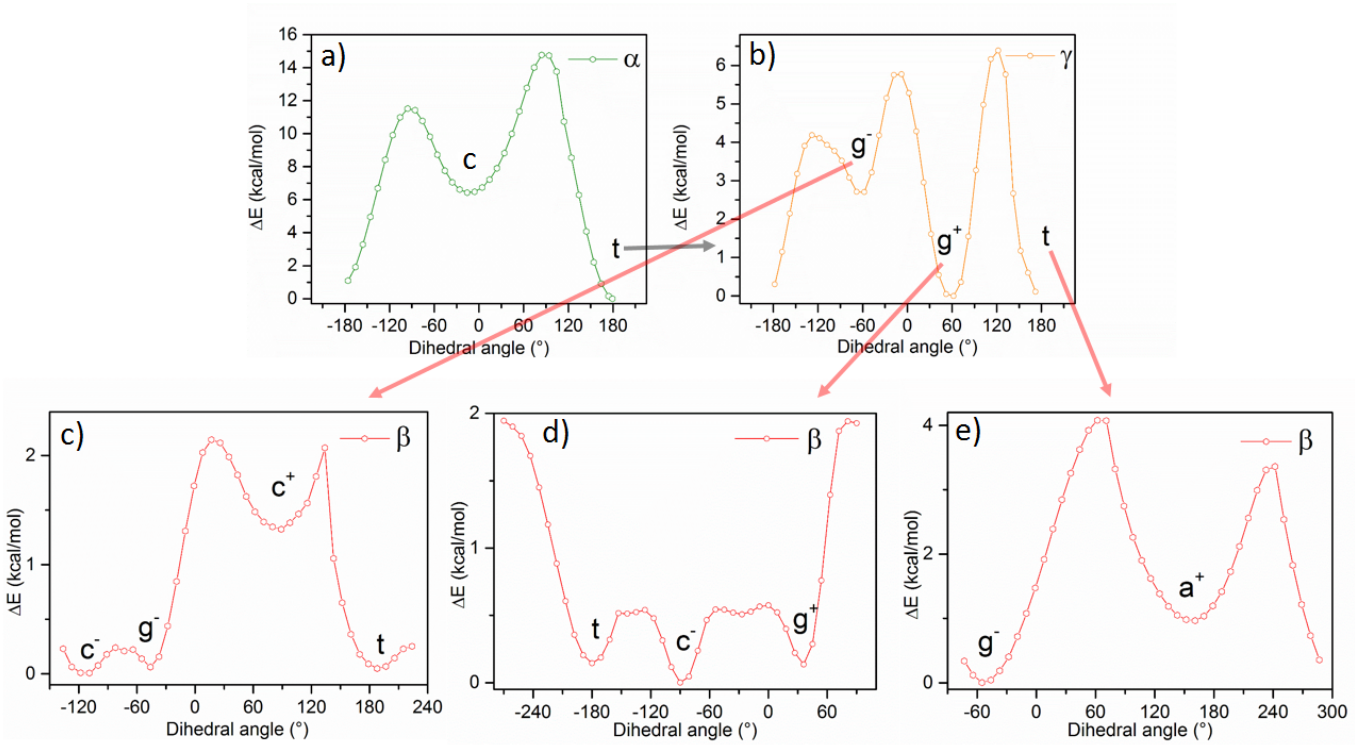


Fig. 6.21: Dihedral angle scans of D-DMT. Arrows show the sequence of calculations. Dihedral angle values which correspond to potential energy minima are labeled by letters.

$-50°$ (labeled g^-) and at $50°$ (labeled a^+), see Figure 6.21, subfigure e). Dihedral angle δ behaved in similar fashion and the potential energy scans are not shown.

From these calculations we obtained a set of dihedral angles of the molecule where the energy of the molecule is lowest. We have built all of the possible conformers of the molecule with the corresponding dihedral angles obtained in the previous scans using the *Gaussview* program [44]. For the middle dihedral angle $\gamma = g^+$ we added dihedral angle $\beta = \delta = r^+$. This dihedral angle was found during to potential energy scan of β when $\gamma = g^-$ and due to the symmetry of the molecule we assumed that it might occur. In total 48 conformations were built.

On these conformers geometry optimization was performed in two different ways: firstly, the dihedral angles containing the hydroxyl group were fixed as shown in Figure 6.20 (circled dihedral angles) and secondly no fixation of dihedral angles was performed during geometry optimization. The reason why we have done this will be described in the following section.

6.4.2 Optimization with fixed dihedral angles

Firstly we have performed geometry optimization on the newly built structures with the dihedral angles containing the hydroxyl group fixed, as shown in Figure 6.20. The reason why we have done this is that we assumed that the hydrogens labeled 19 and 20 will more likely create hydrogen bonding with the solvent as with the O atom of C=O group. We had to force this by fixing these two dihedral angles since the CPCM model does not accurately reflect the interaction of the molecule with the solvent.

For optimization the B3LYP functional and a larger basis set (6-311++G**) was used in order to obtain more accurate results. The CPCM water model was considered for calculations.

15 conformations of lowest enthalpy of the molecule are shown in Table 6.4 (conformers labeled in red color). The highest contribution to the spectra is from conformers with the middle dihedral angle $\gamma = t$ (the first three conformers in Table 6.4). These conformers represent 91% of the molecules which contribute to the spectra. Minor population of conformers with the middle dihedral angle $\gamma = g^+$ is observed (conformers 4-9). Almost negligible populations were calculated for conformers with middle dihedral angle $\gamma = g^-$ (conformers 10-15). In Figure 6.22 images of the most populated conformations of D-DMT for a given dihedral angle γ (t, g^+ and g^-) are shown.

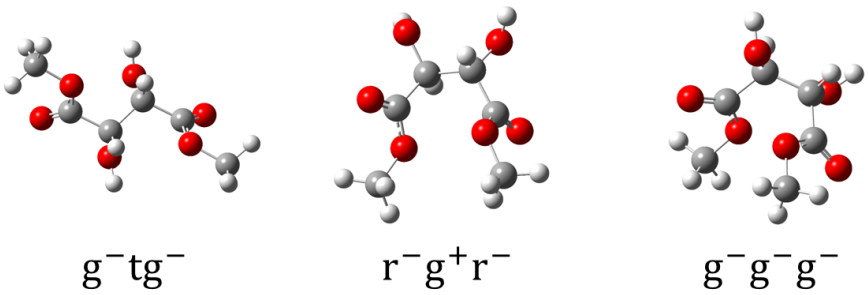


Fig. 6.22: Images of most populated conformations of D-DMT for a given dihedral angle γ (t, g^+ and g^-).

In Figure 6.23 the images of the three most populated conformers among all of the 15 conformers are shown.

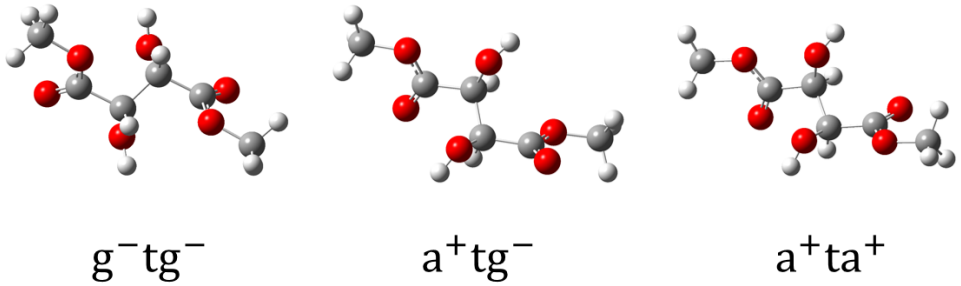


Fig. 6.23: Images of the three most populated conformers among all of the 15 conformers.

6.4.3 Optimization with free dihedral angles

Next, on the structures obtained in the conformation analysis there was a geometry optimization without fixing dihedral angles containing the hydroxyl group performed. We wanted to observe the changes in spectra and conformation in this case in comparison with the fixed dihedral angle optimization.

The B3LYP functional, 6-311++G** basis set and the CPCM water model was used for optimization. Fifteen lowest enthalpy conformations of the molecule are shown in Table 6.4 (conformers labeled blue). Similarly, as it is in the case with fixed dihedral angles, the conformations with the middle dihedral angle $\gamma = \mathrm{t}$ are the most favored. The $\mathrm{g^-tg^-}$ conformer represents 87% of the entire population. This is due to intra-molecular hydrogen bondings which substantially lowers the energy of the molecule. The image of this conformer is shown in Figure 6.24. Hydrogen bonding occurs between oxygen atom number 9 and hydrogen atom number 19, resp. oxygen atom number 12 and hydrogen atom number 20. In both cases the mutual distance between the oxygen and hydrogen atom is 2.1 Å.

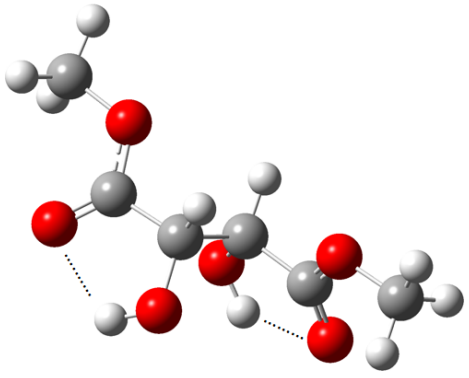


Fig. 6.24: Image of the most populated $\mathrm{g^-tg^-}$ conformer.

However, it seems that the second most favored set of conformers have the middle dihedral angle value $\gamma = \mathrm{g^-}$. This differs from optimization with fixed

dihedral angles, where the second most favored set of conformers has the middle dihedral angle value $\gamma = g^+$. The third most favored set of conformers have the middle dihedral angle $\gamma = g^+$.

Changes in dihedral angles β resp. δ are also observed. These shifts in dihedral angles were expected since optimization without any restrictions allows the molecule to create intra-molecular hydrogen bondings which stabilizes substantially the geometry of the molecule.

Optimizing with fixed and free dihedral angles containing the hydroxyl group represents two different models how to approach the calculation of the spectra. Subsequently, in Section 6.5 we will compare these two models with the experimental spectrum in order to determine which model corresponds better with the experimental results.

Tab. 6.4: This table consists mainly of two parts. Red labeled conformations of the molecule correspond to optimization procedure when the dihedral angles containing the hydroxyl group were fixed during optimization procedure and blue labeled conformations correspond to optimization procedures when the dihedral angles were free. ΔH means the enthalpy difference between conformer no. 1 (i.e. lowest energy conformation of the molecule on the potential energy surface) and the other molecules. By partial population we mean Boltzmann weighted population of the molecule for a given dihedral angle γ (t, g^+ or g^-). By total population we mean total Boltzmann weighted population of the molecule calculated from all of the 15 conformations of the molecule. All of these values were calculated separately for fixed and free circled dihedral angles (see Figure 6.20).

Conf. N°	Label	Dihedral angle (°) α	β	γ	δ	ε	ΔH ($\frac{kcal}{mol}$)	Population (%) Partial	Total
1	g^-tg^-	-179	-55	-179	-57	-179	0.00	64.80	58.95
2	a^+tg^-	179	131	176	-55	-179	0.57	24.70	22.47
3	a^+ta^+	178	134	171	134	178	1.08	10.50	9.55
4	$r^-g^+r^-$	178	-88	53	-92	178	1.72	39.30	3.20
5	tg^+t	-176	-163	58	-154	-176	2.07	21.78	1.78
6	$r^-g^+r^+$	-176	-88	50	84	-179	2.24	16.45	1.34
7	g^+g^+t	177	37	54	-173	-178	2.42	12.07	0.98
8	$r^+g^+r^+$	-179	80	47	91	-179	2.79	6.50	0.53
9	$g^+g^+g^+$	178	27	52	29	177	3.09	3.92	0.32
10	$g^-g^-g^-$	-177	-33	-40	-34	-177	3.36	23.31	0.20
11	tg^-g^-	178	166	-45	-39	-177	3.37	22.92	0.20
12	$r^-g^-r^-$	178	-87	-57	-91	178	3.38	22.34	0.19
13	tg^-g^-t	177	156	-48	158	177	3.61	15.31	0.13
14	$r^-g^-r^+$	178	-90	-61	92	-179	3.81	10.90	0.09
15	$r^+g^-r^+$	-179	89	-64	92	-179	4.25	5.21	0.05
1	g^-tg^-	179	-64	-176	-64	179	0.00	87.00	86.41
2	a^+tg^-	-179	114	173	-59	-179	1.17	12.07	11.98
3	a^+ta^+	177	130	167	114	-179	2.68	0.93	0.93
4	$a^-g^-g^-$	-179	-121	-59	-56	-179	3.22	73.84	0.37
5	$a^-g^-a^+$	-177	-136	-65	136	177	4.20	14.15	0.07
6	tg^-g^-	177	151	-36	-48	-178	4.71	5.97	0.03
7	$r^-g^-r^-$	178	-81	-47	-84	177	4.88	4.53	0.02
8	$r^-g^-r^+$	178	-82	-53	98	-178	5.93	0.76	0.00
9	tg^-t	177	147	-39	147	177	6.02	0.66	0.00
10	$r^+g^-a^+$	180	80	-63	105	-179	7.15	0.10	0.00
11	$g^-g^+g^+$	178	-73	69	63	179	3.85	74.73	0.11
12	$g^+g^+g^-$	178	55	66	-55	-178	4.87	13.52	0.02
13	$r^-g^+r^-$	177	-93	62	-91	177	5.06	9.78	0.02
14	tg^+g^+	178	150	63	46	177	6.12	1.62	0.00
15	tg^+t	-175	-152	57	-174	-176	7.03	0.35	0.00

6.4.4 Calculated VA/VCD and Raman/ROA spectra of D-DMT

From all of the optimized structures shown in Table 6.4 Raman/ROA and VA/VCD spectra have been calculated using the B3LYP functional, 6-311++G** basis set and CPCM water model. In case of Raman/ROA spectra the half width was set to 10 cm^{-1} and in case of VA/VCD spectra the half width was set to 16 cm^{-1}. In case of conformers, when dihedral angles containing the hydroxyl group were fixed, different analysis of calculated spectra were carried.

Firstly, in Figure 6.25 demonstration of Boltzmann weighting is shown. It can be observed that the molecule is very flexible. For example for the middle dihedral angle $\gamma = g^-$ different combinations of dihedral angles β and δ were found. In the Raman spectra a good overlay of spectra is observed in the 1400-1800 cm^{-1} spectral region. In the lower spectral region substantial differences in the spectra are observed. In case of ROA spectra big differences are observed throughout the whole spectral region.

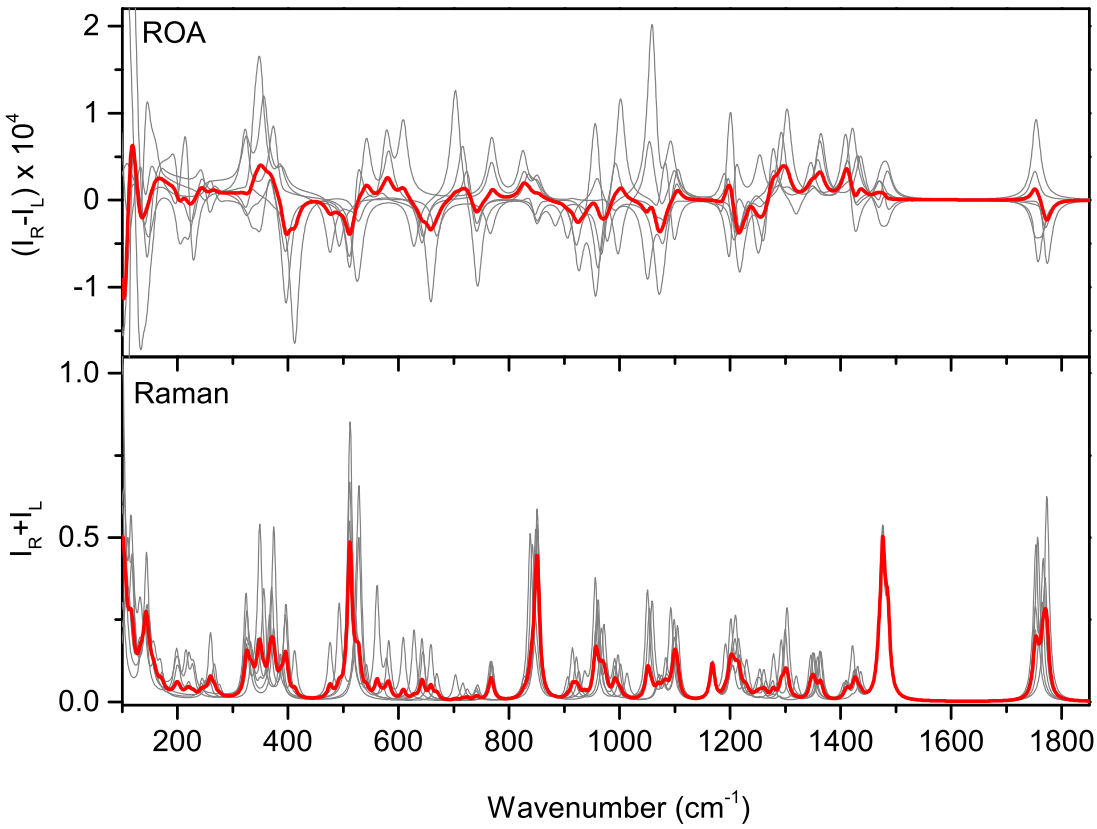


Fig. 6.25: Demonstration of Boltzmann weighting for dihedral angle $\gamma = g^-$. The gray spectra represent all the conformers found for the middle dihedral angle $\gamma = g^-$ and the red line represents Boltzmann weighted average spectrum according to Table 6.4.

Secondly, in Figure 6.26 Raman/ROA spectra for different dihedral angle γ are shown for comparison. More precisely, for dihedral angle labeled $\gamma = \mathrm{t}$, $\gamma = \mathrm{g}^+$ and $\gamma = \mathrm{g}^-$. All spectra are Boltzmann weighted according to Table 6.4. Substantial differences are observed throughout the whole spectral region for both Raman and ROA spectra. This demonstrates the influence of changes in the middle dihedral angle γ on the calculated spectra.

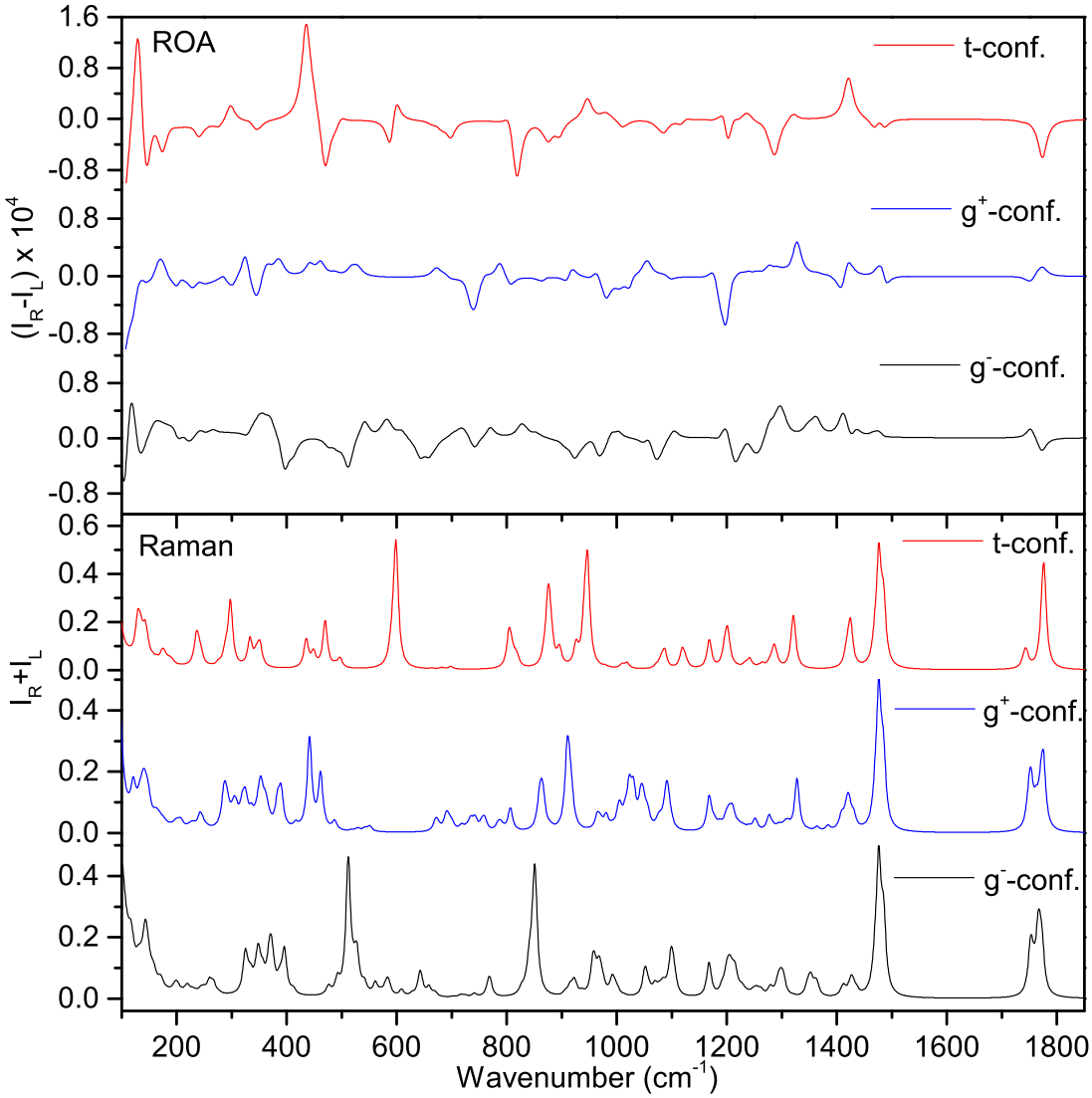


Fig. 6.26: Calculated Raman/ROA spectra of D-DMT for different dihedral angle γ, more precisely for dihedral angle labeled t, g^+ and g^-. All spectra are Boltzmann weighted according to Table 6.4.

Thirdly, in Figure 6.27 there is a demonstration of Boltzmann weighting for the three most populated conformations and for all of the fifteen conformations shown. Almost no difference in the spectra is observed. This means that the conformations with the middle dihedral angle $\gamma = \mathrm{t}$ contribute to the spectra most while the other conformations contribution to the calculated spectra is very little.

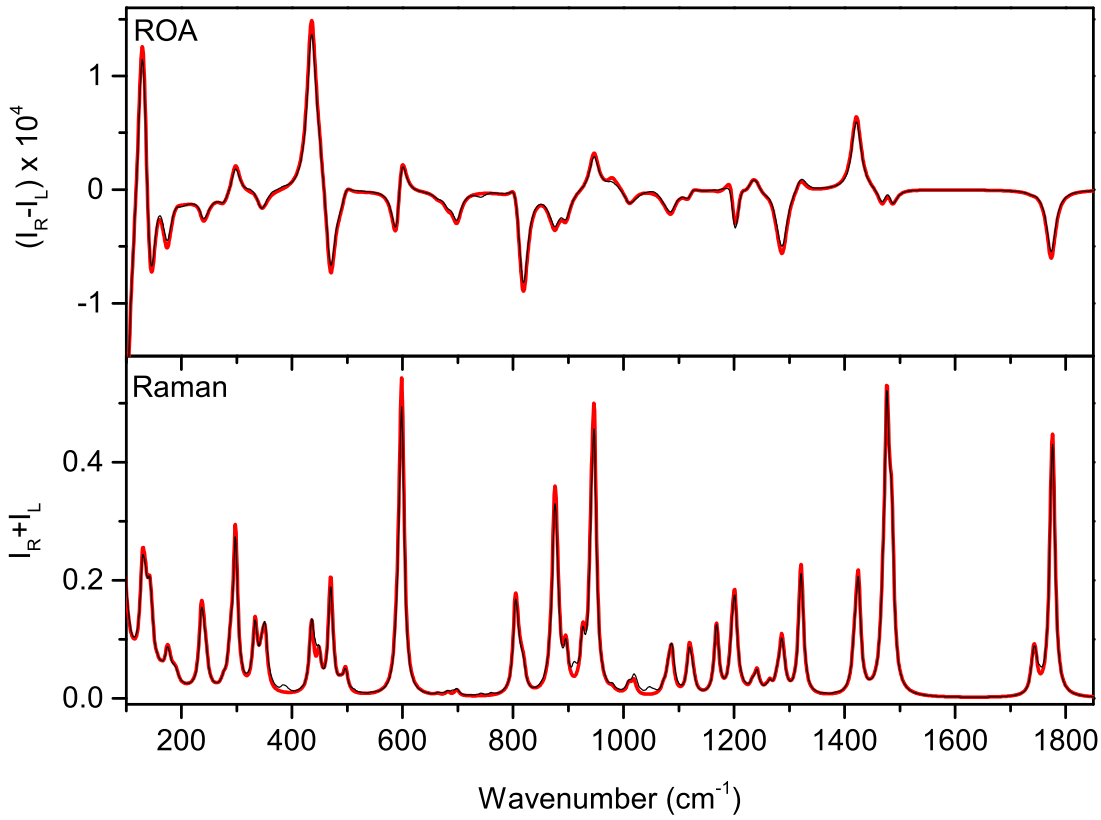


Fig. 6.27: Demonstration of Boltzmann weighting for the three most populated conformations (red spectra) and all of the fifteen conformation (black spectra) according to Table 6.4.

Fourthly, different DFT functionals were tried to optimize the structures obtained in the conformation analysis and to calculate ROA spectra. In Figure 6.28 calculated ROA spectra by using different DFT functionals are compared. All calculations were performed by using the 6-311++G** basis set and the CPCM water model. It is observed that functionals B3LYP and B3PW91 yield almost the same spectra. ROA spectra are shifted to the left and intensity changes are observed when using functional BP86 in comparison with functionals B3PW91 or B3LYP. On the other hand no major differences are observed between the spectra (for example appearance of new spectral bands). Thus it seems that the B3LYP functional represents a reliable computational procedure to perform calculations.

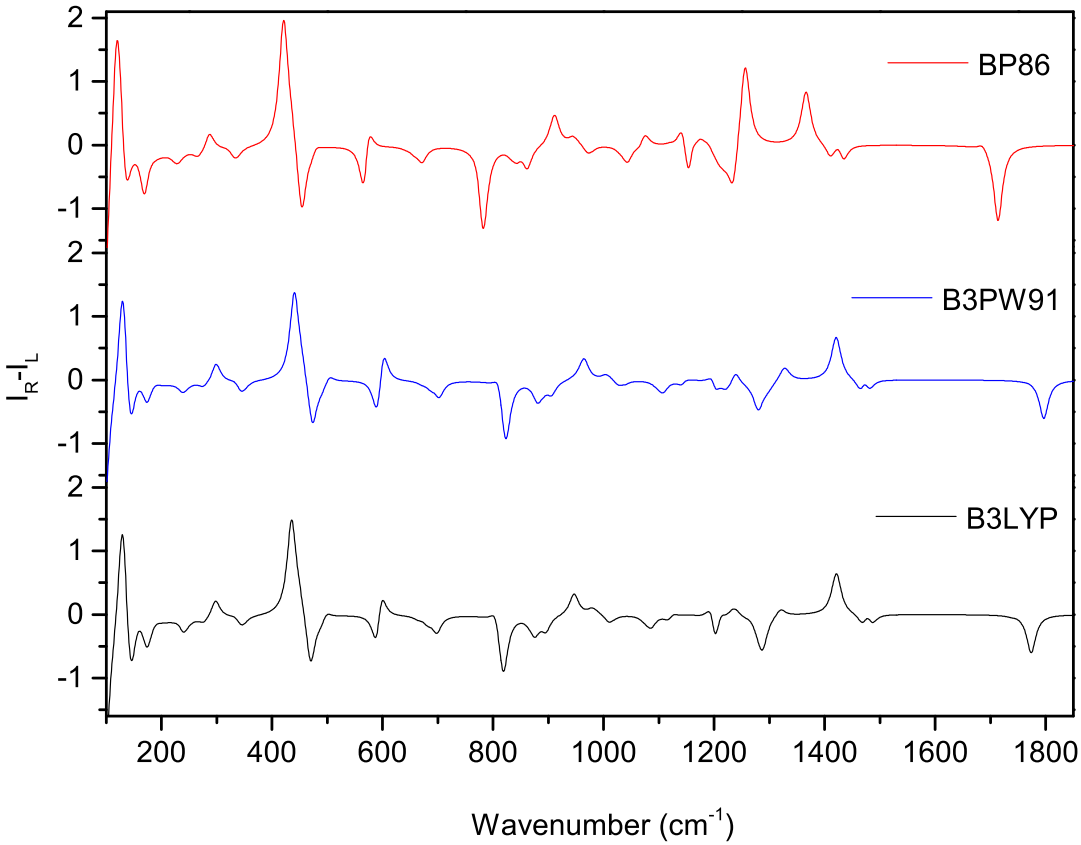


Fig. 6.28: ROA spectra of D-DMT calculated by different variants of DFT functionals. All DFT calculations were performed using 6-311++G** basis set and the CPCM water model.

The final VA/VCD and Raman/ROA spectra with Boltzmann weighting for both fixed and free dihedral angles are shown in Figure 6.29. Differences in spectra are observed. These spectra will be discussed in more detail when comparing them to experimental spectra.

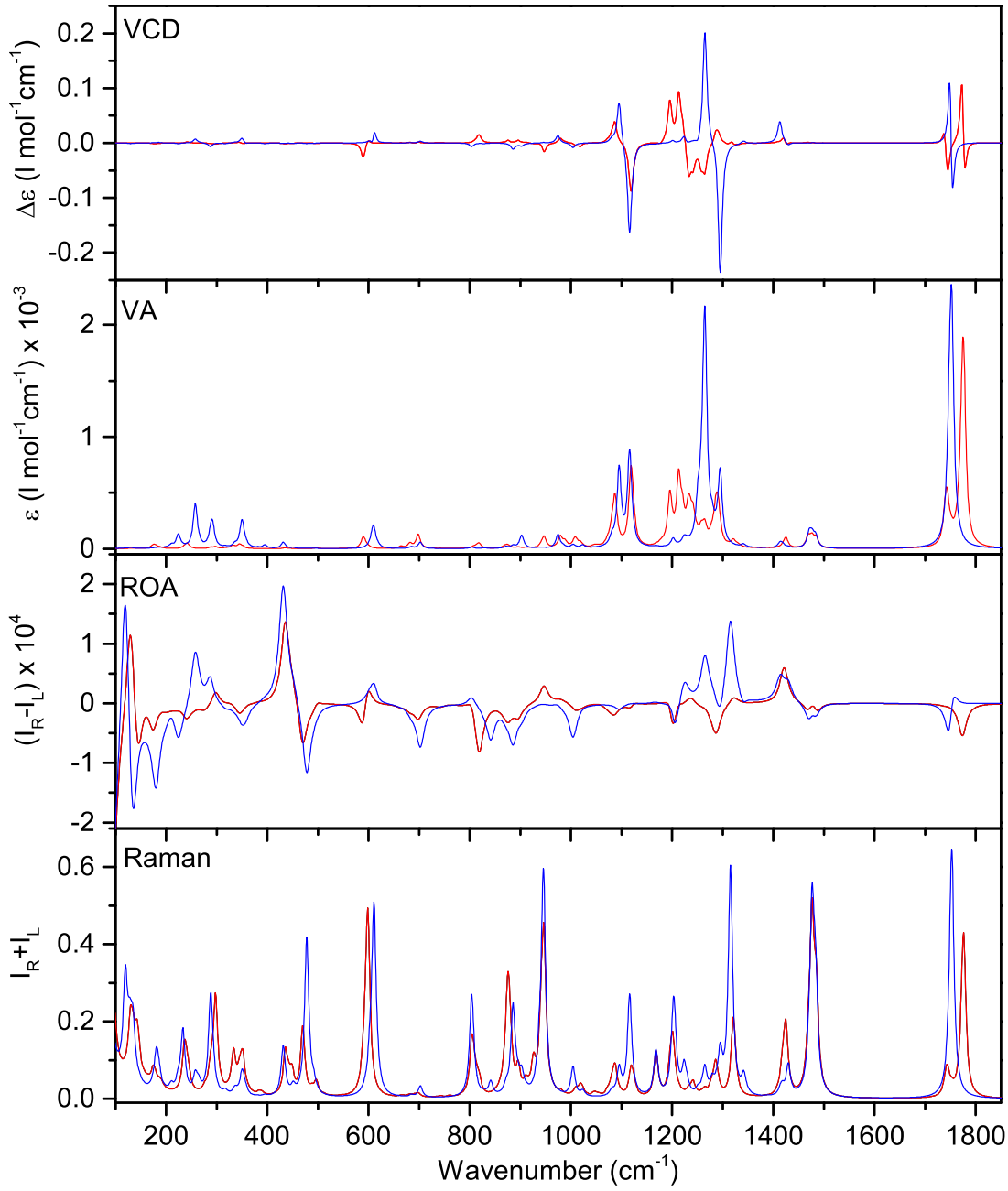


Fig. 6.29: Final Boltzmann weighted VA/VCD and Raman/ROA spectra of D-DMT when dihedral angles containing the hydroxyl group were fixed (red spectra) and free (blue spectra).

6.5 Comparison of experimental and theoretical spectra

In this section we compare the calculated and experimental spectra of D-DMT in H_2O, which was the main goal of this work. In Figure 6.30 the labeled Raman/ROA resp. VA/VCD spectra are compared with their experimental counterparts. The experimental spectra are compared with both theoretical models, i.e. when dihedral angles containing the hydroxyl group were fixed (we will refer to them as to red spectra) resp. when they were left free (we will refer to them as to blue spectra).

In Table 6.5 selected experimental Raman/ROA and VA/VCD bands are assigned to their calculated counterparts. In the same table the selected bands are described by their corresponding vibrational motion which helps us in the interpretation of the experimental spectra. Each vibrational mode was visualized in the *Gaussview* program. Selected vibrational modes were visualized with their corresponding displacement vectors in Figure 6.31 in order to get a better overview of the vibrational motion of D-DMT. In this section, there was different numbering of experimental wavebands applied compared to the numbering in the previous experimental sections, since we labeled only the most intensive bands with reasonable agreement with their calculated counterparts.

Firstly, by comparing the VA/VCD spectra, a better agreement is observed when comparing the experimental spectra with the calculated red spectra. More precisely, by examining the VCD spectrum in the C=O stretching region (approx. 1750 cm^{-1}), a better agreement is observed for the red spectrum. This could possibly suggest that the oxygen atom of the C=O group prefers to create hydrogen bonding with the H_2O compared to intra-molecular hydrogen bondings.

It is worth mentioning that the presence of the band 1c in the VCD spectrum of the C=O stretching region is a confirmation of the presence of the a^+tg^- conformer in the solution.

Secondly, we discuss the region of 1200-1300 cm^{-1}. According to Table 6.5, in this region there are C-O-H resp. O-H deformation vibrations present. Likewise better agreement of the experimental and calculated red spectra is observed. It seems that fixing the hydrogen atoms of the hydroxyl group in a way that they are oriented towards the solvent, reflects the reality better as when they are oriented towards the C=O group. This might be a very good indicator that the hydrogen bonding between the hydrogen atom of the hydroxyl group and the oxygen atom of the H_2O solvent is present.

Similarly, in these regions a better agreement of ROA/Raman spectra is observed between the experimental and red spectra but on the other hand, the low resolution of ROA spectra in this region makes the interpretation difficult.

As stated in the previous sections, ROA and Raman spectra were measured mainly in order to examine spectral region under 1000 cm^{-1}. According to Table 6.5 this region contains skeletal vibrations, in particular. Generally, slightly better agreement is observed between the experimental and blue spectra. It seems that fixing the dihedral angles containing the hydroxyl group affects the skeletal vibrations in a negative way.

Despite the fact that neither the red spectra, nor the blue spectra model describe the conformation perfectly, there are regions where the agreement between the experiment and particular model is quite reasonable. This indicates that the conformers mentioned in Table 6.4 (g^-tg^-, a^+tg^- and a^+ta^+) are likely present in the H_2O solution. On the other hand, besides these conformers there must be additional conformers present that could contribute to better explanation of the experimental results. However, these additional conformers seems to be beyond the limits of models used in this study. As a result, it would be beneficial to incorporate the solvent into the calculations explicitly and see if the modified model works better than the presented ones.

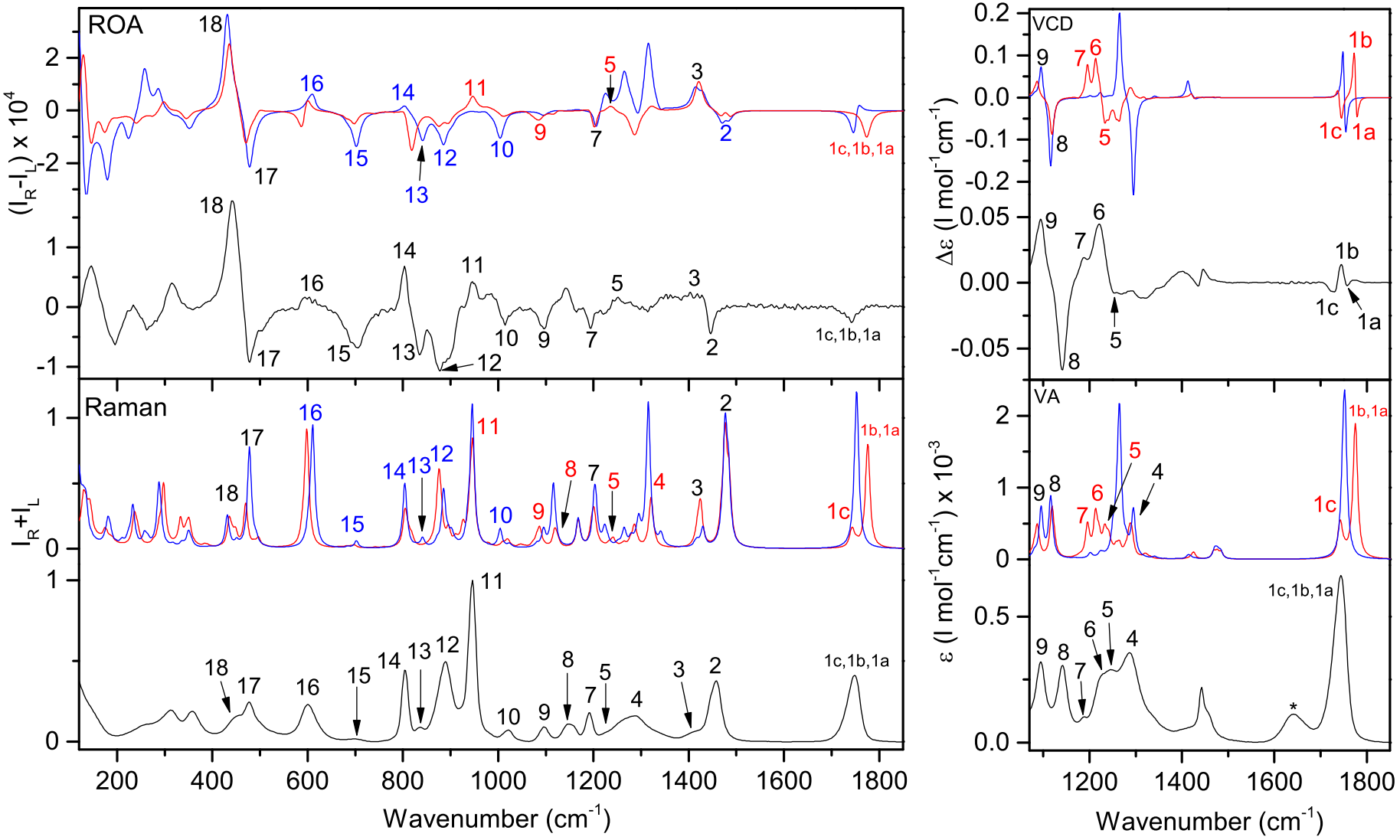


Fig. 6.30: Comparison of Raman/ROA (left figure) and VA/VCD (right figure) spectra of D-DMT with the corresponding experimental spectra in H_2O. Experimental spectra are plotted black, calculated spectra when the dihedral angles containing hydroxyl group were fixed are plotted red and when they were left free they are plotted blue. In case there was better correspondence between the experimental and blue spectra identified, the vibrational band was labeled blue. In case there was better correspondence between the experimental and red spectra identified, the vibrational band was labeled red. When both blue and red spectra corresponded to the experimental spectra, their were labeled black.

Tab. 6.5: Approximate assignment of selected experimental Raman/ROA and VA/VCD bands to their calculated counterparts.

		Frequency (cm^{-1})								
		Raman		ROA		VA		VCD		
mode	symmetry	exp.	calc.	exp.	calc.	exp.	calc.	exp.	calc.	Mode description
1a	A		1777		1777		1777	1756	1778	ν(C=O) (g^-tg^- conf.)
1b	B	1751	1774	1741	1774	1746	1774	1744	1772	ν(C=O) (g^-tg^- conf.)
1c	B		1744		1744		1744	1729	1744	ν(C=O) (a^+tg^- conf.)
2	A	1458	1475	1446	1475	-	-	-	-	$\delta(CH_3)$
3	B	1406	1425	1406	1425	-	-	-	-	δ_{ip}(C*H)
4	A	1291	1292	-	-	1291	1292	-	-	δ(C*H+O-H)
5	A	1225	1233	1222	1233	1245	1233	1250	1233	$\nu(C\text{-}OCH_3)+\delta(CH)+\delta(COH)$
6	A	-	-	-	-	1213	1220	1221	1213	$\nu(C\text{-}OCH_3)+\delta(CH)+\delta(COH)$
7	A	1191	1201	1194	1201	1185	1201	1188	1201	δ(COC*)
8	A	1120	1120	-	-	1142	1120	1142	1120	ν(C*-OH)
9	B	1096	1085	1095	1085	1094	1096	1094	1096	ν(C*-OH)
10	A	1022	1010	1014	1010	-	-	-	-	$\nu(CO\text{-}CH_3)$
11	B	946	945	946	945	-	-	-	-	$\nu(CO\text{-}CH_3)$
12	A	889	885	877	885	-	-	-	-	$\delta_{oop}(OCO)+\delta(CC^*C^*)$
13	B	835	841	835	841	-	-	-	-	$\delta_{oop}(OCO)+\delta(CC^*C^*)$
14	A	803	803	803	803	-	-	-	-	$\delta_{ip}(OCO)+\delta(CC^*C^*)$
15	B	703	702	703	702	-	-	-	-	$\delta_{ip}(OCO)+\delta(CC^*C^*)$
16	A	599	610	599	610	-	-	-	-	$\delta(OCO)+\delta(CC^*C^*)$
17	A	478	477	478	477	-	-	-	-	δ(CC*C*C)
18	B	447	431	447	431	-	-	-	-	δ(CC*C*C)

C* – chiral carbon, ν – stretching vibration, δ – deformation vibration, δ_{ip} – in plane deformation vibration, δ_{oop} – out of plane deformation vibration.

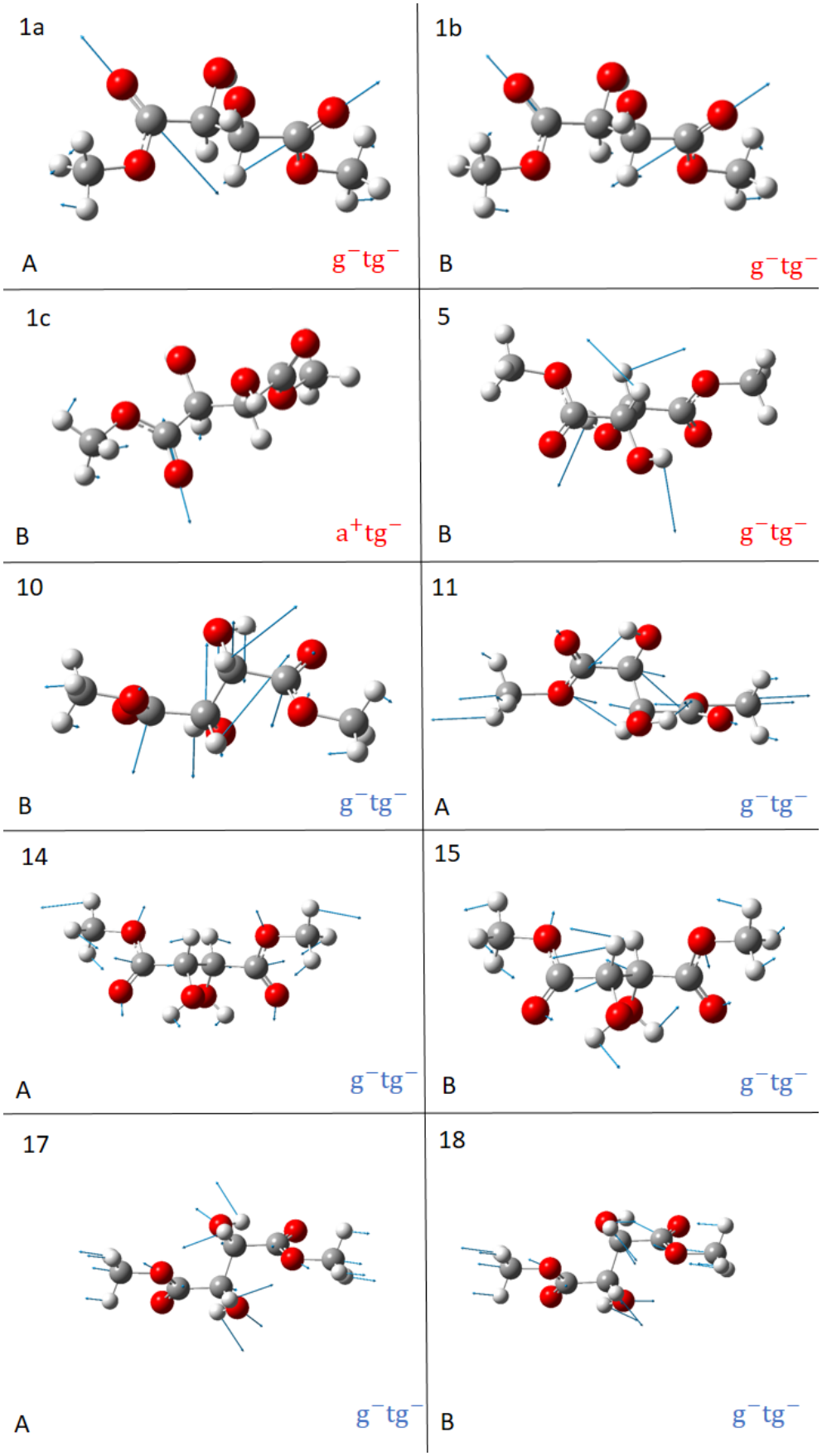


Fig. 6.31: Demonstration of several vibrational modes with corresponding displacement vectors of D-DMT. The number in the top left corner characterizes the vibrational mode based on Table 6.5. The label in the bottom right corner characterizes the conformer responsible for the corresponding vibrational mode based on Table 6.4. The letter in the bottom left corner characterizes the symmetry of the vibration.

7. Conclusions

This thesis was set out to explore the capabilities of VCD and ROA to investigate the conformation of DMT in different solvents. The main research goals and the corresponding results of the thesis are summarized and motivated in the following paragraph:

1. One of the first tasks of this work was to assemble the cuvette used in the Chiral*IR-2X*^{TM} spectrometer so that we obtain VCD spectrum with a baseline as straight as possible. After trying different orientations of the top glass of the cuvette, the position corresponding to the straightest VCD baseline was marked.

2. In the case of VCD, the main motivation was to measure the spectrum in water environment, since this has only been reported in one study [9]. In relation to this challenging task, in this work, first successful measurements of VCD spectra of DMT were carried out in H_2O and D_2O. The next goal was to investigate solvent effects on DMT in different solvents by VA/VCD. It has been shown that the VA and VCD spectra are solvent dependent. We have discussed the reason of various shifts in the VA/VCD spectra of D-DMT in different solvents which helped us to understand the interaction of the solvent with DMT.

3. In the case of ROA, based on the background research, to our knowledge there is only one study available in the literature [26] and moreover with older equipment. In this work, we aimed at studying ROA in H_2O with up-to-date instrumentation in order to obtain more detailed spectra. Our measurements yielded very good quality ROA spectra of DMT. Moreover, we also aimed at investigating solvent effects. Compared to VCD, ROA allowed us to examine spectral region under 1000 cm^{-1}. We have discussed the reason of various shifts in the Raman/ROA spectra of D-DMT in different solvents which helped us to understand the interaction of the solvent with D-DMT. In addition, we have extended the measurements of Raman/ROA spectra of DMT in different solvents in the C-H stretching region, however, only Raman spectrum measured in H_2O solution has shown reasonable results.

4. On the computational side, the main goal of this work (and also of VOA in general) is to determine the conformation of the molecule with the help of theoretical calculations and compare the results of calculations with experimental findings. By *ab initio* calculations the conformations of DMT were studied in multiple works [9, 27], however, in these studies solvents of low polarity were considered. In this work we attempted to simulate the water environment by creating two different computational models: firstly, when dihedral angles containing the hydroxyl groups of D-DMT were fixed during optimization procedures and were oriented towards the solvent and secondly, when they were left free which yielded conformers with intra-molecular hydrogen bondings. In both cases only three major conformations of D-DMT contributed substantially to the

calculated spectra. Subsequently, the experimental and calculated spectra have been compared. Generally, a better agreement between the experiment and the fixed dihedral model was observed in the C=O, O-H and C-O-H bending region, which could possibly indicate hydrogen bonds between DMT and H_2O. In the region of the skeletal vibrations of D-DMT, there was better agreement observed when applying the free dihedral model. Regarding these results it seems that the reality lies between both computational models.

It needs to be considered, that all calculations were performed using the implicit CPCM water model which does not accurately reflect the interaction between water environment and D-DMT. With regard to the model limitations and its possible modification in order to achieve better agreement between experiment and theory, it would be advantageous to incorporate the solvent into the calculations explicitly. This would involve the combination of molecular dynamic simulations with *ab initio* calculations. Development of a modified model would be an interesting challenge for further study.

Considering the further experimentation, it would be interesting to extend the measurments of ROA spectra in the C-H stretching region.

The thesis main outcome lies in detailed analysis of the behaviour of DMT molecule resulting from experimental as well as theoretical findings. Despite of the limitations of the theoretical models used in this work, the results of both theoretical and experimental parts of this study can serve as a good starting point for follow-up studies in this research field.

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List of abbreviations

BO ..... Born-Oppenheimer

CCD ..... Charge-couple device

CID ..... Circular intensity difference

C-PCM ..... Conductor-like polarizable continuum model

DCP ..... Dual circular polarization

DFT ..... Density functional theory

DMT ..... Dimethyl tartrate

D-DMT ..... (2S, 3S)-(−)-dimethyl tartrate

D-PCM ..... Dielectric polarizable continuum model

EOM ..... Electro-optic modulator

HF ..... Hartree-Fock

ICP ..... Incident circular polarization

ICP-CU ..... ROA spectrometer at the Institute of Physics

LoA ..... Level of artifact spectra

LCP ..... Left circularly polarized

L-DMT ..... (2R, 3R)-(+)-dimethyl tartrate

PCM ..... Polarizable continuum model

PES ..... Potential energy surface

RCP ..... Right circularly polarized

ROA ..... Raman optical activity

SCP ..... Scattered circular polarization

SCP-Biotools . Commercial ROA spectrometer Chiral$RAMAN\text{-}2X^{TM}$

TDPT ..... Time dependent perturbation theory

VA ..... Vibrational absorption

VCD ..... Vibrational cricular dichroism

VOA ..... Vibrational optical activity

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